



A novel self-paced Takagi-Sugeno-Kang fuzzy classifier via learning the robust antecedents of all fuzzy rules

Jia Qu^a, Wulin Yang^a, Zekang Bian^{b,c,*}

^a Department of Computer Science and Artificial Intelligence, Changzhou University, Changzhou, Jiangsu, China

^b Department of AI & Computer Science, Jiangnan University, Wuxi, Jiangsu, China

^c PRC Ministry of Education Engineering Research Center of Intelligent Technology for Healthcare, Wuxi, Jiangsu, China

ARTICLE INFO

Keywords:

Self-paced TSK fuzzy classifier
Ambiguous samples
Fuzzy ensemble clustering
Robust and interpretable antecedents
Fewer major fuzzy rules

ABSTRACT

Due to its high interpretability and superior nonlinear approximation capability, the Takagi-Sugeno-Kang (TSK) fuzzy classifier has been successfully applied to machine learning. However, when handling complex datasets containing ambiguous samples, maintaining or enhancing the generalization capability and interpretability of the TSK fuzzy classifier requires addressing two issues: (1) robust and interpretable antecedents must be established, and (2) ambiguous samples must be adaptively identified. Therefore, inspired by self-paced learning, a novel self-paced Takagi-Sugeno-Kang (TSK) fuzzy classifier via learning the robust antecedents of all fuzzy rules (SP-TSK-RA) is proposed. Specifically, to obtain robust and interpretable antecedents, SP-TSK-RA first introduces a fuzzy ensemble clustering method for complex datasets. Then, a new consequent parameter learning objective function is formulated by introducing self-paced learning to adaptively identify ambiguous samples and enhance the generalization capability on complex data. Additionally, to maintain generalization capability and simultaneously achieve higher interpretability, SP-TSK-RA introduces the L_1 regularization into the consequent parameter learning objective function to explore fewer major fuzzy rules. Experimental results on multiple datasets demonstrate that compared with comparative methods, SP-TSK-RA achieves robust generalization capability and fewer fuzzy rules.

1. Introduction

Classification analysis, a fundamental task in machine learning, aims to automatically predict the labels of unlabeled samples (testing samples) using a classifier trained on labeled samples (training samples). Owing to its superior nonlinear approximation and high interpretability, the Takagi-Sugeno-Kang (TSK) fuzzy classifier [1,2] has attracted increasing attention in recent years.

Unlike common classifiers (e.g., the K-nearest neighbors (KNN) classifier [3]) that operate on the original feature space, the TSK fuzzy classifier maps the original feature space into a fuzzy space, subsequently performing classification via fuzzy logic operators. In

List of Acronyms: TSK, Takagi-Sugeno-Kang; SP-TSK-RA, Self-Paced TSK fuzzy classifier via the robust antecedents of all fuzzy rules; FCM, Fuzzy C-Means; SVM, Support Vector Machine; RF, Random Forest; KNN, K-Nearest Neighbors; LLM, Least Learning Machine; B-TSK-FC, Bayesian TSK Fuzzy Classifier; SVR, Support Vector Regression; GWN, Gaussian White Noise; WL-TSK, Wide Learning based TSK fuzzy classifier; MCMC, Markov Chain Monte Carlo; MH, Metropolis-Hastings; RBF, Radial Basis Function.

* Corresponding author at: Department of AI & Computer Science, Jiangnan University, Wuxi, Jiangsu, China.

E-mail addresses: tb17060015b4@cumt.edu.cn (J. Qu), 18837676108@163.com (W. Yang), zekang_bian@jiangnan.edu.cn (Z. Bian).

<https://doi.org/10.1016/j.ins.2026.123902>

Received 12 March 2026; Received in revised form 12 July 2026; Accepted 13 July 2026

Available online 15 July 2026

0020-0255/© 2026 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

the training stage, the fuzzy classifier fuzzifies the original feature space of the training samples by partitioning the features into numeric fuzzy sets using a membership function [4], thereby constructing the antecedents of the fuzzy rules. After defuzzifying the outputs of the fuzzy rules on the training samples, an objective function is optimized to learn the corresponding consequents of all fuzzy rules. In the test stage, the final labels of test samples are predicted by integrating the defuzzified outputs across all fuzzy rules.

Recently, many TSK-based fuzzy classifiers have been developed to achieve enhanced generalization capability and/or better interpretability for different tasks. Among them, some methods aim to develop single TSK fuzzy models. For example, Cui et al. [5] proposed a minibatch gradient descent algorithm integrated with uniform regularization and batch normalization to speed up TSK training on large-scale and/or high-dimensional datasets. Cao et al. [6] developed a lightweight classifier to utilize adaptive weighting to construct quantitative equivalent fuzzy rules, thereby enhancing the input–output mapping capability. Zhou et al. [7] proposed a fully interpretable first-order system (FIMG-TSK) equipped with negative entropic and rule-stability regularization to simultaneously minimize output variance and ensure rule readability. Lou et al. [8] designed a heterogeneous feature transfer learning framework (HeFT-FIR) to optimize TSK parameters by mining hidden discriminant information across source and target domains. Similarly, Li et al. [9] proposed a multi-view transfer representation learning method (MIP-TRL-FS) for EEG analysis, which preserves information at both sample and feature levels while maintaining the linguistic interpretability of the TSK model. On the other hand, ensemble TSK fuzzy classifiers have been developed to achieve better classification performance. For example, Guo et al. [10] proposed an ensemble classifier integrating improved bagging and dropout strategies to handle high-dimensional problems, constructing concise fuzzy rule sets by aggregating multiple sub-models. Zhou et al. [11] developed a strategy based on internal and generative decorrelation to construct an ensemble of regressors (EFIMG-TSKs) with a quintuple diversity guarantee. Li et al. [12] developed an output stability metric to quantify output variance, effectively solving the optimization problem in the linear aggregation learning of interpretable fuzzy subsystems from a probabilistic perspective. Zhou et al. [13] proposed a novel framework (MV-FIMG-TSKs) by assigning an interpretable TSK sub-classifier to each view and fusing them with accompanying Gaussian Mixture Models to maximize information utilization. Gu et al. [14] developed a fast-training algorithm for adversarial deep fuzzy classifiers, which employs gradient-guided rule downsizing to accelerate the training of stacked sub-classifiers. Although these methods have achieved remarkable success, further improvements are still needed to simultaneously ensure robust generalization and clear interpretability in complex scenarios. In this paper, we focus on developing an efficient single TSK fuzzy classifier to achieve robust generalization and better interpretability. To this end, existing TSK fuzzy classifiers have been developed from the following aspects:

1. Methods for constructing the antecedents of fuzzy rules, with the goal of generating interpretable antecedent centers. On one hand, fixed antecedent centers are generated by hard partitioning methods [15]. For example, the grid-based partitioning method [16] is used to partition the input space into fixed grids, which often leads to the “curse of dimensionality”. To circumvent rule explosion, one can randomly draw fixed antecedent centers from a set of predefined candidates [17,18]. On the other hand, flexible antecedent centers are generated by soft partitioning methods [19]. For example, owing to its simple implementation, fuzzy C-means (FCM) clustering method is often used to generate antecedent centers [20]. To enhance adaptability to complex data structures, the adaptive clustering methods (e.g., DBSCAN) are used to adaptively generate the flexible antecedent centers. In addition, genetic algorithms [21] have been used to optimize the membership function parameters to stabilize antecedent generation.
2. New objective functions for learning the consequents of fuzzy rules. For example, the least squares (LS) method [22] is often employed to learn consequents by minimizing prediction errors. To enhance robustness, regularization terms [23] are introduced into the objective function to achieve enhanced generalization capability. To achieve faster running speed, the least learning machine (LLM) is utilized to learn consequent parameters in one step [24]. By integrating the large margin criterion into the consequent learning objective function, a convex quadratic programming problem is formulated to minimize structural risk and maximize classification margins, thereby enhancing the generalization capability, particularly for high-dimensional multi-label datasets [25].

However, when handling complex datasets containing ambiguous samples, existing TSK-based methods face two critical issues in maintaining generalization capability and interpretability: (i) how to obtain stable antecedent centers, and (ii) how to identify ambiguous samples and circumvent their adverse impact on classification performance.

To address issue (i), this paper develops a fuzzy ensemble clustering method to generate stable and interpretable antecedent centers for all fuzzy rules. For issue (ii), this paper proposes a new consequent parameter learning framework that integrates self-paced learning and L_1 regularization. This framework adaptively identifies ambiguous samples and distills major fuzzy rules, driven by the following motivations:

First, according to [17], the antecedent centers of fuzzy rules are usually obtained using FCM. However, the clustering performance of FCM is sensitive to the initialization, and since traditional FCM relies heavily on minimizing a distance-based objective function, it is easily interfered with by noisy samples. This easily traps the objective function into local optimization and causes antecedent center shifts, especially for complex datasets, which leads to the obtained antecedent centers becoming unstable, thereby making the interpretability unstable and even affecting the fitting capability of the fuzzy rules. Therefore, obtaining stable antecedent centers by learning a stable clustering result has become critical to the antecedent generation process. While some studies [26] show that stable antecedent centers can be achieved by averaging multiple FCM runs, this paper develops a more robust ensemble clustering method. Inspired by ensemble/consensus clustering [27,28], our approach directly generates a stable and unified clustering result. Compared to averaging multiple FCM runs, ensemble clustering aims to reveal the consistent structure of data with superior stability, thereby enhancing the ambiguity handling capability for complex datasets.

Second, due to its high computational efficiency and clear mathematical theory, the least squares method [29] is usually adopted to

learn the consequents of fuzzy rules, but it exhibits high sensitivity to ambiguous samples and inability to compress the fuzzy rules, where these ambiguous samples are always defined as the hard-to-classify samples in high-overlap boundary regions or noisy samples, thereby weakening the performance of the TSK fuzzy classifier. Therefore, according to [30], the self-paced learning strategy learns the model in a “from easy to difficult” manner, providing a flexible mechanism to dynamically evaluate the difficulty of each sample. Since noisy or ambiguous samples are treated as the most difficult samples, they are assigned relatively small weights. Additionally, according to [6], all fuzzy rules should be treated unequally, and the major fuzzy rules should be obtained to improve the interpretability of the TSK-based fuzzy classifiers. Therefore, this paper designs a novel framework to efficiently learn robust consequents and extract major rules by introducing a self-paced learning mechanism and L_1 regularization, respectively, thereby achieving enhanced and robust generalization capability and higher interpretability.

Therefore, inspired by the above considerations, a novel self-paced TSK fuzzy classifier via learning the robust antecedents of all fuzzy rules (SP-TSK-RA) is proposed in this paper. Specifically, different from the traditional TSK fuzzy classifier, SP-TSK-RA constructs stable and interpretable antecedents using the proposed ensemble fuzzy clustering method. Then, SP-TSK-RA proposes a new consequent parameter framework to adaptively identify the weights of ambiguous samples, minimizing their impact on the trained TSK fuzzy classifier by introducing self-paced learning mechanism, enhancing the generalization capability. In addition, major fuzzy rules are distilled during the consequent parameter learning process to achieve better interpretability. The main contributions of this paper are summarized as follows.

- (1) Compared with the traditional TSK fuzzy classifier, SP-TSK-RA constructs robust interpretable antecedents by employing an ensemble clustering method to obtain stable antecedent centers, thereby enhancing the robustness of SP-TSK-RA on complex datasets containing ambiguous samples.
- (2) To adaptively identify ambiguous samples, SP-TSK-RA proposes a new consequent parameter learning framework by introducing a self-paced mechanism, which enhances generalization capability and follows the “from easy to complex” human cognitive behavior.
- (3) Compared with the traditional TSK fuzzy classifier, SP-TSK-RA extracts major fuzzy rules by introducing sparse regularization into the consequent parameter learning framework, which further improves interpretability.
- (4) Experimental results on all utilized datasets demonstrate that SP-TSK-RA achieves more robust generalization capability and better interpretability (e.g., fewer fuzzy rules) than all adopted comparative methods.

The remainder of this paper is organized as follows: In Section 2, the classic TSK fuzzy classifier and related works are briefly reviewed. In Section 3, the details of the proposed method SP-TSK-RA are formulated. In Section 4, comprehensive experiments evaluating SP-TSK-RA on all adopted datasets are presented. In Section 5, conclusions and further work are outlined.

2. Takagi-Sugeno-Kang fuzzy classifier

As a frequently utilized fuzzy classifier, the Takagi-Sugeno-Kang (TSK) fuzzy classifier can achieve enhanced nonlinear fitting capability and linguistic-based interpretability using K fuzzy rules, where the k -th ($k = 1, 2, \dots, K$) fuzzy rule is defined as follows:

$$\begin{aligned} \text{IF } \mathbf{x}_1 \text{ is } A_1^k \wedge \dots \wedge \mathbf{x}_d \text{ is } A_d^k, \\ \text{THEN } y = f^k(\mathbf{x}). \end{aligned} \quad (1)$$

where $\mathbf{x} = [x_1, x_2, \dots, x_d] \in \mathbf{R}^{1 \times d}$ and x_j ($j = 1, 2, \dots, d$) represent a sample and its j -th input feature, d is the number of features. A_j^k denotes the fuzzy subset for the j -th feature in the k -th rule, $\wedge$ represents the fuzzy conjunction operator, and $f^k(\mathbf{x})$ is the output of the k -th fuzzy rule. Clearly, the output of the TSK fuzzy classifier is a linear function, when $f^k(\mathbf{x}) = p_0^k + p_1^k x_1 + \dots + p_d^k x_d$, where $\mathbf{p}^k = [p_0^k, p_1^k, \dots, p_d^k]$ is consequent parameter vector, such that the fuzzy classifier is the first-order TSK fuzzy classifier.

After applying the defuzzification process, the final output $y(\mathbf{x})$ of the first-order TSK fuzzy classifier can be denoted as follows:

$$y(\mathbf{x}) = \frac{\sum_{k=1}^K \mu^k(\mathbf{x}) f^k(\mathbf{x})}{\sum_{k=1}^K \mu^k(\mathbf{x})} = \sum_{k=1}^K \tilde{\mu}^k(\mathbf{x}) f^k(\mathbf{x}) \quad (2)$$

where $\mu^k(\mathbf{x})$ represents the firing level of $\mathbf{x}$ for the k -th fuzzy rule, $\tilde{\mu}^k(\mathbf{x})$ denotes the normalized firing level. They are defined by:

$$\mu^k(\mathbf{x}) = \prod_{j=1}^d u_j^k(x_j) \quad (3)$$

$$\tilde{\mu}^k(\mathbf{x}) = \frac{\mu^k(\mathbf{x})}{\sum_{k=1}^K \mu^k(\mathbf{x})} \quad (4)$$

where $u_j^k(x_j)$ is generally implemented utilizing Gaussian function:

$$u_j^k(x_j) = \exp\left(-\frac{(x_j - c_j^k)^2}{2\sigma_j^k}\right) \quad (5)$$

where c_j^k and σ_j^k represent the j -th element in the center vector $\mathbf{c}^k = [c_1^k, \dots, c_d^k]$ and width vector $\sigma^k = [\sigma_1^k, \dots, \sigma_d^k]$ of the k -th fuzzy rule, respectively. Generally, $\mathbf{c}^k$ can be generated by a fuzzy clustering method or randomly selected from five fixed centers [31], and σ_j^k can be calculated by:

$$\sigma_j^k = h \frac{\sum_{i=1}^n u_{ik} (x_{ji} - c_j^k)^2}{\sum_{i=1}^n u_{ik}} \quad (6)$$

where h is the preset parameter, n denotes the overall count of input samples. u_{ik} represents the membership degree of the i -th input sample $\mathbf{x}_i$ belonging to the k -th cluster, and x_{ji} represents the j -th feature of the i -th input sample $\mathbf{x}_i$. After $\mathbf{c}^k$ and σ^k are obtained, the antecedents of the K fuzzy rules are determined. Then, as introduced earlier, several methods (e.g., LLM) can be employed to learn the consequents of all fuzzy rules. Mathematically, the output of the first-order TSK fuzzy classifier is formulated as:

$$y(\mathbf{x}) = \mathbf{p}_g^T \mathbf{x}_g \quad (7)$$

where $\mathbf{x}_g$ and $\mathbf{p}_g$ are defined as follows:

$$\mathbf{x}_e = (1, \mathbf{x}^T)^T \quad (8)$$

$$\tilde{\mathbf{x}}^k = \tilde{\mu}^k(\mathbf{x}) \mathbf{x}_e \quad (9)$$

$$\mathbf{x}_g = \left((\tilde{\mathbf{x}}^1)^T, (\tilde{\mathbf{x}}^2)^T, \dots, (\tilde{\mathbf{x}}^K)^T \right)^T \quad (10)$$

$$\mathbf{p}^k = (p_0^k, p_1^k, \dots, p_d^k)^T \quad (11)$$

$$\mathbf{p}_g = \left((\mathbf{p}^1)^T, (\mathbf{p}^2)^T, \dots, (\mathbf{p}^K)^T \right)^T \quad (12)$$

The first-order TSK fuzzy classifier is inherently suited for addressing classification tasks. To transform the continuous outputs of the TSK fuzzy classifier into discrete class labels, considering computational efficiency and simple implementation, according to [17,32], the shortest distance-based method is adopted in this paper. Although the shortest distance-based method may artificially impose an ordinal structure on nominal datasets, unlike traditional global linear regression models, the TSK fuzzy classifier treats class labels as independent coordinate targets and precisely approximates them through the activation of different local rule combinations [33], thereby effectively circumventing the influence of the artificially imposed ordinal structure. Specifically, the label of a test sample $\mathbf{x}$ can be predicted easily: for binary classification tasks (e.g., with class labels $\{-1, 1\}$), by taking the sign of the model's output on $\mathbf{x}$; for multiple classification tasks, the target label is assigned by finding the category with the smallest absolute deviation from the continuous predicted value. Formally, suppose the discrete label set of a C -class classification task is $L = \{1, 2, \dots, C\}$. Given a continuous predicted value $\hat{y}(\mathbf{x})$, the final predicted class label $\hat{y}_{final}$ is obtained by $\hat{y}_{final} = \arg\min_{c \in L} |\hat{y}(\mathbf{x}) - c|$.

3. A novel self-paced Takagi-Sugeno-Kang fuzzy classifier via learning the robust antecedents of all fuzzy rules

In this section, we detail the proposed method SP-TSK-RA, and its flowchart is illustrated in Fig. 1.

As shown in Fig. 1, in the training process, SP-TSK-RA first adopts the proposed ensemble clustering method to obtain the stable antecedent centers of all fuzzy rules. Then, SP-TSK-RA employs a consequent parameter learning framework by introducing self-paced learning to adaptively identify the ambiguous samples and reduce their impact on classification performance, thereby enhancing generalization capability. In addition, SP-TSK-RA can adopt the major fuzzy rules to obtain robust generalization capability alongside better interpretability. Subsequently, in the test process, for the test data, SP-TSK-RA first calculates the firing levels using the learned stable antecedent parameters. Then, it computes the continuous prediction value via the distilled major fuzzy rules, and obtains the final output class label using the shortest distance-based method.

Subsequently, we elaborate on the specific mechanisms of the proposed method SP-TSK-RA, including: (1) the training procedure of SP-TSK-RA, and (2) the test procedure of SP-TSK-RA.

3.1. Training procedure of SP-TSK-RA

3.1.1. The construction process of antecedents of all fuzzy rules

Unlike a single fuzzy C-means clustering method (FCM) used to obtain the centers of the Gaussian-based membership function in the TSK fuzzy classifier [32], to achieve enhanced and stable antecedents of all fuzzy rules for complex datasets, the proposed method

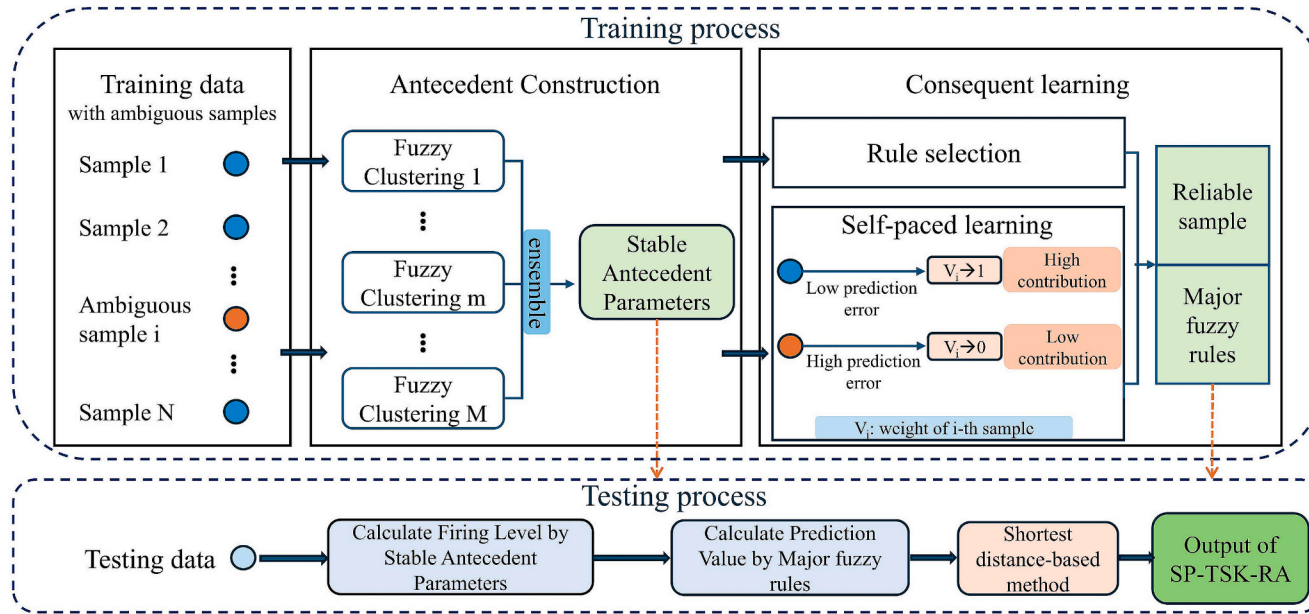


Fig. 1. The framework of SP-TSK-RA. During the training process, stable antecedents are first constructed via fuzzy ensemble clustering, and then the consequent learning is executed to identify ambiguous samples and select major rules. Then learned antecedent and consequent parameters are transferred to the testing process for the label prediction.

SP-TSK-RA proposes a fuzzy ensemble clustering method by integrating several base clustering results, where all base clustering results are generated by calling FCM with different random initializations, thereby reducing the instability of the single clustering result on the antecedent centers of all fuzzy rules. The objective function of the proposed fuzzy ensemble clustering method is formulated as:

$$\begin{aligned}
 J_{\text{ensemble}}\left(u_{ik}^m, \mathbf{v}_k^m, \bar{u}_{ik}, \tilde{w}_m\right) &= \sum_{m=1}^M \tilde{w}_m \sum_{i=1}^N \sum_{k=1}^K \left(u_{ik}^m\right)^q \left\|\mathbf{x}_i - \mathbf{v}_k^m\right\|^2 \\
 &+ \alpha \sum_{m=1}^M \sum_{i=1}^N \sum_{k=1}^K \bar{u}_{ik} \ln\left(\frac{\bar{u}_{ik}}{u_{ik}^m}\right) + \beta \sum_{m=1}^M \tilde{w}_m \ln \tilde{w}_m, \\
 \text{s.t. } \sum_{m=1}^M \tilde{w}_m &= 1, \quad 0 \leq \tilde{w}_m \leq 1, \quad \sum_{k=1}^K u_{ik}^m = 1, \quad \bar{u}_{ik} = \frac{1}{M} \sum_{m=1}^M u_{ik}^m.
 \end{aligned} \tag{13}$$

where M , N , and K are the total number of the base clustering results, samples, and fuzzy rules, respectively. q is the fuzzy index, set as $q = 2$, $\mathbf{v}_k^m$ and u_{ik}^m represent the k -th cluster center and the membership degree of the i -th sample belonging to the k -th cluster in the m -th base clustering result, respectively. $\tilde{w}_m$ is the weight of the m -th base clustering result and is adaptively adjusted. $\bar{u}_{ik}$ is the ensemble clustering result. Parameters α and β control the contributions of the corresponding terms.

Obviously, each term of Eq.(13) can be described as follows: the first term $\sum_{i=1}^N \sum_{k=1}^K \left(u_{ik}^m\right)^q \left\|\mathbf{x}_i - \mathbf{v}_k^m\right\|^2$ of Eq.(13) represents the m -th base clustering result by running FCM. The second term $\sum_{m=1}^M \sum_{i=1}^N \sum_{k=1}^K \bar{u}_{ik} \ln\left(\frac{\bar{u}_{ik}}{u_{ik}^m}\right)$ of Eq.(13) forces the membership degree u_{ik}^m of each classifier to approximate the ensemble clustering result $\bar{u}_{ik}$. The third term $\sum_{m=1}^M \tilde{w}_m \ln \tilde{w}_m$ of Eq.(13) is the information entropy-based regularization term, which adaptively assigns weights to M base clustering results.

After optimizing Eq.(13), we can obtain the optimal $\mathbf{v}_k^m$ and $\tilde{w}_m$ ($m = 1, 2, \dots, M$ and $k = 1, 2, \dots, K$), and then we can obtain the ensemble and stable antecedent centers $\mathbf{v}_k$ ($k = 1, 2, \dots, K$) of K fuzzy rules by:

$$\mathbf{v}_k = \frac{\sum_{m=1}^M \tilde{w}_m \mathbf{v}_k^m}{\sum_{m=1}^M \tilde{w}_m}. \tag{14}$$

Then, we can further obtain $\sigma^k = [\sigma_1^k, \dots, \sigma_d^k]$, where σ_j^k ($k = 1, 2, \dots, K$ and $j = 1, 2, \dots, d$) is calculated by:

$$\sigma_j^k = h \frac{\sum_{i=1}^N \bar{u}_{ik} (\mathbf{x}_{ij} - \mathbf{v}_i^k)^2}{\sum_{i=1}^N \bar{u}_{ik}} \tag{15}$$

where d is the number of features, and h is the preset parameter.

To optimize the proposed objective function, we adopt the alternating optimization method, and the details are described below.

Step 1: Update u_{ik}^m with fixed $\mathbf{v}_k^m$, $\bar{u}_{ik}$, and $\tilde{w}_m$.

When $\mathbf{v}_k^m$, $\bar{u}_{ik}$, and $\tilde{w}_m$ are fixed, Eq.(13) becomes:

$$\begin{aligned}
 J_{\text{ensemble}}\left(u_{ik}^m\right) &= \tilde{w}_m \sum_{i=1}^N \sum_{k=1}^K \left(u_{ik}^m\right)^q \left\|\mathbf{x}_i - \mathbf{v}_k^m\right\|^2 \\
 &+ \alpha \sum_{m=1}^M \sum_{i=1}^N \sum_{k=1}^K \bar{u}_{ik} \ln\left(\frac{\bar{u}_{ik}}{u_{ik}^m}\right), \\
 \text{s.t. } \sum_{k=1}^K u_{ik}^m &= 1, \quad \bar{u}_{ik} = \frac{1}{M} \sum_{m=1}^M u_{ik}^m.
 \end{aligned} \tag{16}$$

Inspired by [34], we adopt an approximate solution method to update u_{ik}^m . By treating $\bar{u}_{ik}$ as a fixed constant in the current iteration and initially ignore the regularization term, then we can derive the standard FCM membership degree update formula:

$$\tilde{u}_{ik}^m = \frac{1}{\sum_{l=1}^K \left(\frac{\left\|\mathbf{x}_i - \mathbf{v}_k^m\right\|}{\left\|\mathbf{x}_i - \mathbf{v}_l^m\right\|}\right)^{\frac{2}{q-1}}}. \tag{17}$$

Subsequently, by introducing regularization correction, the weighted mixing adjustment is adopted to obtain $\hat{u}_{ik}^m$ by:

$$\hat{u}_{ik}^m = (1 - \theta) \tilde{u}_{ik}^m + \theta \bar{u}_{ik}, \tag{18}$$

where θ is a mixing coefficient, i.e., $\theta = \frac{\alpha}{\alpha+1}$. So we can get the final update formula of u_{ik}^m as follows:

$$u_{ik}^m = \frac{(1 - \theta)\tilde{u}_{ik}^m + \theta\bar{u}_{ik}}{\sum_{l=1}^K \left((1 - \theta)\tilde{u}_{il}^m + \theta\bar{u}_{il} \right)}. \quad (19)$$

Step 2: Update $\mathbf{v}_k^m$ with fixed u_{ik}^m , $\bar{u}_{ik}$, and $\tilde{w}_m$.

When u_{ik}^m , $\bar{u}_{ik}$, and $\tilde{w}_m$ are fixed, Eq.(13) becomes:

$$J_{\text{ensemble}}(\mathbf{v}_k^m) = \sum_{m=1}^M \tilde{w}_m \sum_{i=1}^N \sum_{k=1}^K (u_{ik}^m)^q \|\mathbf{x}_i - \mathbf{v}_k^m\|^2, \quad (20)$$

Let $\frac{\partial J_{\text{ensemble}}(\mathbf{v}_k^m)}{\partial \mathbf{v}_k^m} = 0$, we get:

$$\mathbf{v}_k^m = \frac{\sum_{i=1}^N (u_{ik}^m)^q \mathbf{x}_i}{\sum_{i=1}^N (u_{ik}^m)^q} \quad (21)$$

Step 3: Update $\bar{u}_{ik}$ with fixed u_{ik}^m , $\mathbf{v}_k^m$, and $\tilde{w}_m$.

According to the condition in Eq.(13), we can directly update $\bar{u}_{ik}$:

$$\bar{u}_{ik} = \frac{1}{M} \sum_{m=1}^M u_{ik}^m \quad (22)$$

Step 4: Update $\tilde{w}_m$ with fixed u_{ik}^m , $\mathbf{v}_k^m$, and $\bar{u}_{ik}$.

When u_{ik}^m , $\mathbf{v}_k^m$, and $\bar{u}_{ik}$ are fixed, Eq.(13) becomes:

$$J_{\text{ensemble}}(\tilde{w}_m) = \sum_{m=1}^M \tilde{w}_m \sum_{i=1}^N \sum_{k=1}^K (u_{ik}^m)^q \|\mathbf{x}_i - \mathbf{v}_k^m\|^2 + \beta \sum_{m=1}^M \tilde{w}_m \ln \tilde{w}_m, \quad (23)$$

$$\text{s.t. } \sum_{m=1}^M \tilde{w}_m = 1, \quad 0 \leq \tilde{w}_m \leq 1.$$

Let $J_m = \sum_{i=1}^N \sum_{k=1}^K (u_{ik}^m)^q \|\mathbf{x}_i - \mathbf{v}_k^m\|^2$, to solve this constrained optimization problem, we construct the Lagrangian function of Eq. (23) by introducing the Lagrangian multiplier δ :

$$\mathcal{L}(\tilde{\mathbf{w}}_m, \delta) = \sum_{m=1}^M \tilde{w}_m J_m + \beta \sum_{m=1}^M \tilde{w}_m \ln \tilde{w}_m + \delta \left(\sum_{m=1}^M \tilde{w}_m - 1 \right). \quad (24)$$

After taking the partial derivative with respect to $\tilde{w}_m$ and setting it to zero, we can get:

$$\tilde{w}_m = \frac{e^{-\frac{J_m}{\beta}}}{\sum_{m=1}^M e^{-\frac{J_m}{\beta}}} \quad (25)$$

The details of optimizing Eq.(13) can be summarized in Algorithm 1.

Algorithm 1. The details of optimizing Eq.(13)

Input: Base data $\{\mathbf{x}_i\}$, $i = 1, 2, \dots, N$, number of clusters K , number of clustering algorithms M , fuzzy index q , the regularization parameters α, β .

- 1 **Initialize:** Using the standard FCM formula to generate u_{ik}^m , randomly generate M sets of initial clustering centers $\mathbf{v}_k^m$, initialize global consensus membership degree $\bar{u}_{ik}$ and the weights $\tilde{w}_m = 1/M$.
- 2 **Repeat**
- 3 Update u_{ik}^m by (19).
- 4 Update $\mathbf{v}_k^m$ by (21).
- 5 Update $\bar{u}_{ik}$ by (22).
- 6 Update $\tilde{w}_m$ by (25).
- 7 **Until** $\max_i |(u_{ik}^m)^{(t)} - (u_{ik}^m)^{(t-1)}| < 1e-4$
- end while**
- 8 **Output:** The optimal $\mathbf{v}_k^m$ and $\tilde{w}_m$.

After calling Algorithm 1, we can obtain the optimal $\mathbf{v}_k^m$ and $\tilde{w}_m$, then we can calculate the ensemble and stable antecedent centers $\mathbf{v}_k$ ($k = 1, 2, \dots, K$) and the corresponding widths σ^k ($k = 1, 2, \dots, K$) of K fuzzy rules using Eq.(14) and Eq.(15), respectively.

3.1.2. The learning process of consequent parameters of all fuzzy rules

After generating stable antecedent centers and widths through the aforementioned methods, to mitigate the rule explosion problem [4], we assign a rule weight w_k to each rule and apply L_1 regularization to assign these weights. This means the model automatically selects the rules that contribute most significantly to classification, eliminating unimportant or redundant rules, which further

improves interpretability. The objective function of the proposed framework can be formulated as follows:

$$\min_{\mathbf{w}, \mathbf{P}} \sum_{i=1}^N \left(y_i - \sum_{k=1}^K w_k \tilde{\mu}_k(\mathbf{x}_i) \mathbf{p}_k^T \tilde{\mathbf{x}}_i \right)^2 + \gamma \|\mathbf{w}\|_1, \quad (26)$$

$$\text{s.t. } 0 \leq w_k \leq 1, \quad k = [1, 2, \dots, K].$$

where N and K denote the overall count of samples and fuzzy rules, respectively. $\mathbf{x}_i$ denotes the i -th sample, and $\tilde{\mathbf{x}}_i = (1, \mathbf{x}_i)$, y_i represents the label of $\mathbf{x}_i$, $\tilde{\mu}_k(\mathbf{x}_i)$ denotes the membership degree of $\mathbf{x}_i$ for the k -th fuzzy rule, $\mathbf{p}_k^T$ represents the consequent parameters of the k -th fuzzy rule. And $\mathbf{w} = [w_1, w_2, \dots, w_K]^T \in \mathbf{R}^K$ represents the weight vector of K fuzzy rules. Parameter γ is used to adjust the strictness of rule selection.

To circumvent the impact of ambiguous samples on the classification performance of TSK fuzzy classifier, we attempt to identify these ambiguous samples and then reduce their influence on classification performance. Inspired by self-paced learning [35], during the learning process, an increasing number of reliable samples are adopted to train the classifier, and the classification performance of the classifier is gradually enhanced. In this way, reliable samples are assigned higher weights, and unreliable samples (ambiguous samples) are assigned lower weights. Thus, Eq.(26) can be transformed into:

$$\min_{\mathbf{v}, \mathbf{w}, \mathbf{P}} \sum_{i=1}^N v_i \left(y_i - \sum_{k=1}^K w_k \tilde{\mu}_k(\mathbf{x}_i) \mathbf{p}_k^T \tilde{\mathbf{x}}_i \right)^2 + \lambda \sum_{i=1}^N (1 - v_i)^2 + \gamma \|\mathbf{w}\|_1, \quad (27)$$

$$\text{s.t. } 0 \leq v_i \leq 1, \quad i = [1, 2, \dots, N], 0 \leq w_k \leq 1, \quad k = [1, 2, \dots, K].$$

where $\lambda \sum_{i=1}^N (1 - v_i)^2$ is the adaptive self-paced learning term, and $\mathbf{v} = [v_1, v_2, \dots, v_N]^T \in \mathbf{R}^N$ represents the weight vector of N samples. The hyperparameter λ is used to control the progress of self-paced learning. In practice, we initialize λ with a small value and then gradually increase it during the optimization process.

Obviously, during the optimization process of Eq.(27), ambiguous samples can be automatically identified, assigned relatively small weights. Furthermore, to enhance generalization capability and robustness, inspired by [36], the commonly used term, e.g., L_2 regularization, is introduced into Eq.(27), and the new objective function can be defined as follows:

$$\min_{\mathbf{v}, \mathbf{w}, \mathbf{P}} \sum_{i=1}^N v_i \left(y_i - \sum_{k=1}^K w_k \tilde{\mu}_k(\mathbf{x}_i) \mathbf{p}_k^T \tilde{\mathbf{x}}_i \right)^2 + \lambda \sum_{i=1}^N (1 - v_i)^2 + \gamma \|\mathbf{w}\|_1 + \eta \|\mathbf{P}\|_F^2, \quad (28)$$

$$\text{s.t. } 0 \leq v_i \leq 1, i = [1, 2, \dots, N], 0 \leq w_k \leq 1, k = [1, 2, \dots, K].$$

Two hyperparameters γ and η are used to control the contributions of the corresponding terms. Here we have some discussion about the objective function in Eq.(28). It can adaptively identify ambiguous samples by observing the difference between the predicted labels and the true labels. If the difference is large, the sample is considered ambiguous and assigned a lower weight, and a smaller difference yields a higher weight. Moreover, owing to the incorporation of L_1 and L_2 regularization, the model can maintain good generalization capability and robustness with fewer rules.

In this paper, due to easy implementation, the block coordinate descent method [37] is adopted to optimize the objective function in Eq.(28).

Step 1: Update $\mathbf{v}$ with $\mathbf{w}$ and $\mathbf{P}$ fixed.

When $\mathbf{w}$ and $\mathbf{P}$ are fixed, Eq.(28) becomes:

$$\min_{0 \leq v_i \leq 1} \sum_{i=1}^N v_i \left(y_i - \sum_{k=1}^K w_k \tilde{\mu}_k(\mathbf{x}_i) \mathbf{p}_k^T \tilde{\mathbf{x}}_i \right)^2 + \lambda \sum_{i=1}^N (1 - v_i)^2 \quad (29)$$

Let $e_i = y_i - \sum_{k=1}^K w_k \tilde{\mu}_k(\mathbf{x}_i) \mathbf{p}_k^T \tilde{\mathbf{x}}_i$, taking the partial derivative of the objective function with respect to v_i and setting it to zero, we have:

$$e_i^2 - 2\lambda(1 - v_i) = 0. \quad (30)$$

Solving this equation yields $v_i = 1 - \frac{e_i^2}{2\lambda}$. Considering the constraint $0 \leq v_i \leq 1$, we project the solution into the feasible region and obtain its closed-form solution:

$$v_i = \max\left(0, 1 - \frac{e_i^2}{2\lambda}\right) \quad (31)$$

where e_i denotes the difference between the predicted value and the true label value for the i -th sample. As mentioned above, we can assess sample reliability by observing the value of e_i .

Step 2: Update $\mathbf{w}$ with $\mathbf{v}$ and $\mathbf{P}$ fixed.

Similar to the optimization of $\mathbf{v}$, when $\mathbf{v}$ and $\mathbf{P}$ are fixed, Eq.(28) becomes:

$$\min_{0 \leq \mathbf{w} \leq 1} \sum_{i=1}^N v_i \left(y_i - \sum_{k=1}^K w_k \tilde{\mu}_k(\mathbf{x}_i) \mathbf{p}_k^T \tilde{\mathbf{x}}_i \right)^2 + \gamma \|\mathbf{w}\|_1 \quad (32)$$

We split the summation term into two parts: $j = k$ and $j \neq k$. By setting the partial derivative with respect to w_k to zero, we obtain its closed-form solution:

$$w_k = \frac{\sum_{i=1}^N v_i c_i d_{i,k} - \frac{\gamma}{2}}{\sum_{i=1}^N v_i d_{i,k}^2} \quad (33)$$

where $d_{i,k} = \tilde{\mu}_k(\mathbf{x}_i) \mathbf{p}_k^T \tilde{\mathbf{x}}_i$, $c_i = y_i - \sum_{j \neq k} w_j \tilde{\mu}_j(\mathbf{x}_i) \mathbf{p}_j^T \tilde{\mathbf{x}}_i$. Considering the constraints applied to w_k , we can get:

$$w_k = \min \left(\max \left(\frac{\sum_{i=1}^N v_i c_i d_{i,k} - \frac{\gamma}{2}}{\sum_{i=1}^N v_i d_{i,k}^2}, 0 \right), 1 \right) \quad (34)$$

From Eq.(34), we can obtain the major fuzzy rules, which further improves interpretability.

Step 3: Update $\mathbf{P}$ with $\mathbf{v}$ and $\mathbf{w}$ fixed.

When $\mathbf{w}$ and $\mathbf{v}$ are fixed, Eq.(28) becomes:

$$\min_{\mathbf{P}} \sum_{i=1}^N v_i \left(y_i - \sum_{k=1}^K w_k \tilde{\mu}_k(\mathbf{x}_i) \mathbf{p}_k^T \tilde{\mathbf{x}}_i \right)^2 + \eta \|\mathbf{P}\|_F^2 \quad (35)$$

Similar to the previous steps, we split the summation term with respect to $\mathbf{p}_k$ into two parts: $j = k$ and $j \neq k$. Let $b_i^{(k)} = y_i - \sum_{j \neq k} w_j \tilde{\mu}_j(\mathbf{x}_i) \mathbf{p}_j^T \tilde{\mathbf{x}}_i$. By setting the gradient with respect to the vector $\mathbf{p}_k$ to zero, we obtain:

$$-2 \sum_{i=1}^N v_i w_k \tilde{\mu}_k(\mathbf{x}_i) \mathbf{x}_i \left(b_i^{(k)} - w_k \tilde{\mu}_k(\mathbf{x}_i) \mathbf{x}_i^T \mathbf{p}_k \right) + 2\eta \mathbf{p}_k = 0. \quad (36)$$

By rearranging the above equation, we can obtain its closed-form solution:

$$\mathbf{p}_k = (\mathbf{A}_k + \eta \mathbf{I})^{-1} \mathbf{b}_k, \quad (37)$$

where the matrix $\mathbf{A}_k$ and vector $\mathbf{b}_k$ are explicitly defined as:

$$\mathbf{A}_k = \sum_{i=1}^N v_i w_k^2 \tilde{\mu}_k^2(\mathbf{x}_i) \tilde{\mathbf{x}}_i \tilde{\mathbf{x}}_i^T \quad (38)$$

$$\mathbf{b}_k = \sum_{i=1}^N v_i w_k \tilde{\mu}_k(\mathbf{x}_i) \tilde{\mathbf{x}}_i \left(y_i - \sum_{j \neq k} w_j \tilde{\mu}_j(\mathbf{x}_i) \mathbf{p}_j^T \tilde{\mathbf{x}}_i \right) \quad (39)$$

where $\mathbf{I}$ is the identity matrix.

The details of optimizing Eq.(28) are summarized in Algorithm 2.

Algorithm 2. The optimization process of Eq.(28)

Input: All Training samples $\mathbf{X} \in \mathbf{R}^{N \times d}$, the number of fuzzy rules K , the maximum value of the hyperparameter λ , the regularization parameters γ and η .
1 Initialize: Using the least squares method to generate the consequent parameters $\mathbf{P}$, randomly generate $\mathbf{v}$ and $\mathbf{w}$.
2 Repeat
 3 Update $\mathbf{v}$ by Eq.(31).
 4 Update $\mathbf{w}$ by Eq.(34).
 5 Update $\mathbf{P}$ by Eq.(37).
 6 Update $\lambda \leftarrow \lambda \times 1.1$
 7 **Until** training loss $\tau \leq 1e-4$
end while
8 Output: The optimal $\mathbf{p}_k$ and w_k .

3.2. Test procedure of SP-TSK-RA

After the training procedure of SP-TSK-RA, we can obtain the optimal $\mathbf{p}_k$ and w_k . Then we can calculate the label prediction value $\hat{y}$ for a test sample $\mathbf{x}'$, and the output $\hat{y}$ of SP-TSK-RA is defined as:

$$\hat{y} = \sum_{k=1}^K w_k \tilde{\mu}_k(\mathbf{x}') \mathbf{p}_k^T \tilde{\mathbf{x}}, \quad (40)$$

where $\tilde{\mathbf{x}} = [1, \mathbf{x}']$, and $\tilde{\mu}_k(\mathbf{x}')$ is calculated by Eq.(4). Then, $\hat{y}$ can be further transformed into the predicted label according to the

shortest distance-based method. The details of the test procedure of SP-TSK-RA can be outlined in Algorithm 3.

Algorithm 3. The testing procedure of SP-TSK-RA

Input: Testing samples $\mathbf{X}' \in \mathbf{R}^{n \times d}$, optimal $\mathbf{v}_k$ and σ^k , optimal consequent parameters $\mathbf{p}_k$ and w_k .
 1 For each testing sample $\mathbf{x}'_i, i = 1, 2, \dots, n$.
 2 Obtain $\tilde{\mu}_k(\mathbf{x}'_i)$ by Eq.(4).
 3 Obtain prediction value $\hat{y}(\mathbf{x}'_i)$ by Eq.(40), then obtain the final label $\hat{y}_{final}(\mathbf{x}'_i)$ by taking shortest distance-based method.
 4 End for.
 5 **Output:** The predicted label set $\mathbf{Y}$.

3.3. Time and model complexity analysis

In this subsection, the time and model complexity of SP-TSK-RA will be analyzed, i.e., the training procedure and test procedure.

The training procedure of SP-TSK-RA depends on Algorithm 1 and Algorithm 2. According to Algorithm 1, steps 3, 4, 5, and 6 need $O(MNKd)$, $O(MNKd)$, $O(NMK)$, and $O(MNKd)$ computational costs, respectively, where M , N , K , and d are the total number of base clustering results, all training samples, fuzzy rules, and features, respectively.

Therefore, the overall computational burden of Algorithm 1 is $O(I_1KNMd)$, where I_1 is the maximum number of iterations in Algorithm 1. According to Algorithm 2, steps 3, 4, and 5 need $O(NK(d+1))$, $O(NK^2(d+1))$, and $O(KN(d+1)^2 + K(d+1)^3 + K^2N(d+1))$, respectively. Therefore, the overall computational burden of Algorithm 2 is about $O(I_2(KNd^2 + K^2Nd + Kd^3))$, where I_2 is the maximum number of iterations in Algorithm 2. In summary, the time complexity of the training procedure of SP-TSK-RA is $O(I_2K(Nd^2 + KNd + d^3) + I_1KNMd)$.

The test procedure of SP-TSK-RA depends on Algorithm 3. Specifically, steps 2 and 3 need $O(nKd)$ and $O(nKd)$ computational costs, respectively, where n is the total number of all test samples. Therefore, the time complexity of the test procedure of SP-TSK-RA is $O(nKd)$. Clearly, the test phase is time-efficient. Furthermore, due to the incorporated rule selection design, K is smaller, leading to further improved test efficiency.

In summary, the overall time complexity of SP-TSK-RA is $O(I_2K(Nd^2 + KNd + d^3) + I_1KNMd + nKd)$.

Next, let us analyze the model complexity (MC) of SP-TSK-RA and other TSK-based fuzzy classifiers in terms of both the number of adopted fuzzy rules and adopted features. Suppose the dataset has d original features and the number of adopted fuzzy rules is K . For the first-order-based TSK fuzzy classifiers (i.e., L2-TSK, First TSK, and SP-TSK-RA), their IF-parts contain $2Kd$ parameters and THEN-parts contain $K(d+1)$ parameters. Thus, their model complexities are $MC = K(3d+1)$. For the zero-order-based TSK fuzzy classifiers (i.e., Zero TSK, B-TSK-FC, and WL-TSK), their THEN-parts only contain K consequent parameters. According to [1], their model complexities can be evaluated as $MC = K(2d+1)$.

4. Experiments

In this section, we evaluate the efficiency of the proposed method SP-TSK-RA by comparing its performance (i.e., classification performance, interpretability, and running times) with that of all comparative methods. All experiments are carried out on a computer with an Intel(R) Core(TM) i5-13400 16-Core Processor CPU equipped with 16 GB of memory.

Table 1
Details of 20 adopted datasets.

Datasets	Number of samples	Number of features	Number of classes
Banknote(BAN)	1372	4	2
Blood(BLO)	748	4	2
Dermatology(DER)	366	34	6
Heart(HEA)	270	13	2
ILPD(ILP)	583	10	2
Balance(BAL)	625	4	3
Mammographic(MAM)	961	5	2
Magic(MAG)	19,020	10	2
Page(PAG)	5473	10	5
Ecoli (ECO)	336	7	8
Contraceptive(CON)	1473	9	3
TAE	151	5	3
Vertebrate 2C(V2C)	310	6	2
Auto mpg(AUT)	398	7	3
Vertebrate 3C (V3C)	310	6	3
Waveform1-2-vs-3(WAV)	2746	21	2
Fertility(FER)	100	9	2
Plrx(PLR)	182	12	2
Shuttle-subset(SST)	2157	9	7
ZOO	101	16	7

4.1. Experiment organization

4.1.1. Datasets and comparative methods

Within our experiments, twenty datasets acquired from the UCI repository are adopted. Their sizes range from 100 to 19020, and feature dimensions range from 4 to 34. The details of all adopted datasets are reported in Table 1.

To evaluate the efficiency of SP-TSK-RA, five TSK-based methods (i.e., Zero-order TSK fuzzy classifier [22], First-order TSK fuzzy classifier [38], L2-TSK fuzzy classifier [39], B-TSK-FC [40], WL-TSK [1]) are adopted in our experiments. Since the TSK fuzzy classifier inherently functions as a regression-based model that maps continuous outputs to discrete class labels. Similarly, SVR [41], as a regression variant of the classic and effective Support Vector Machine, shares a similar regression-based classification mechanism. Therefore, we adopt SVR as a non-fuzzy comparative method to further verify the performance of the proposed method. Furthermore, to provide a broader and more comprehensive performance evaluation, K-nearest neighbors (KNN) [3] is also introduced as an additional classic non-fuzzy baseline. All comparative methods are briefly introduced as follows:

Zero-order TSK fuzzy classifier [22]: The classifier employs constant consequents in fuzzy rules, facilitating simple defuzzification for nonlinear function approximation.

First-order TSK fuzzy classifier [38]: The classifier employs linear functions of inputs as consequents, together with the zero-order TSK, it serves as a baseline for comparing other fuzzy classifiers.

L2-TSK fuzzy classifier [39]: The classifier incorporates L_2 regularization into the consequent learning objective to penalize large parameters, promoting robustness against overfitting.

B-TSK-FC [40]: It integrates the TSK fuzzy classifier into a Bayesian inference framework, proposing a novel fuzzy classifier. This method constructs the classifier by simultaneously learning the antecedent and consequent parameters of fuzzy rules, significantly improving interpretability compared to the classic TSK fuzzy classifier.

WL-TSK [1]: It is a novel wide TSK fuzzy classifier that integrates the concept of biological plausibility with wide learning architectures. The algorithm introduces a “fuzzy-knowledge-out” mechanism to construct diverse sub-classifiers, which are ultimately merged into a single zero-order TSK classifier that retains enhanced generalization capability and interpretability.

SVR [41]: The classifier is a variant of SVM that performs classification by combining regression outputs with a threshold.

KNN [3]: The classifier is a classic instance-based learning algorithm that predicts the class label of a target sample based on the majority vote of its k nearest neighbors in the feature space.

4.1.2. Experimental settings

In our experiments, all adopted hyperparameters of SP-TSK-RA are set as follows:

In the training procedure, during the construction process of antecedents of all fuzzy rules, four hyperparameters (M , q , α and β) are set. According to [42], we set $M = 4$ and $q = 2$, we set α by grid search from $\{0.001, 0.01, 0.05, 0.1, 0.5, 1, 5\}$, and set β by grid search from $\{0.1, 1, 5, 10, 100\}$. During the learning process of consequent parameters, four hyperparameters (K , λ , γ and η) need to be set. According to [31], K is tuned over the interval $[c, 12c]$ using a grid search strategy with an increment of $\text{round}(c/2)$, in which c stands for the class count and $\text{round}(\bullet)$ represents the round value of $\bullet$. Because the hyperparameter λ has a key impact on the efficiency of the self-paced learning, according to [37], we initialize λ to 0.1 and gradually increase it by 10% in each iteration, setting its maximum value by grid search from $\{0.2, 0.5, 1, 5, 10\}$. For parameters γ and η , we set them by grid search from $\{0.001, 0.01, 0.05, 0.1, 0.5, 1, 1.5, 5\}$. As for First-TSK and Zero-TSK, following [31], their rule quantities are set via grid search traversing the ranges $[c, 12c]$ and $[c, 20c]$ respectively, both utilizing a uniform step length of $\text{round}(c/2)$. For L2-TSK, following [40], we set the regularization parameter β by grid search from $\{0.125, 0.25, 1, 2, 8, 32, 128\}$, the search range for its fuzzy rules is the same as First-TSK.

For B-TSK-FC, according to [40], we set K by grid search from $\{2, 3, \dots, 15\}$ for binary classification tasks, and span the interval $[c, 10c]$ with increments of $\text{round}(c/2)$ for multiple classification tasks. Three parameters α , C , and h are set by grid search from $\{0.1, 0.5, 1, 1.5, 2\}$, $\{2^{-3}, 2^{-1}, 2^0, \dots, 2^5, 2^7\}$, and $\{0.16, 1, 2.56\}$, respectively.

For WL-TSK, according to [1], too few rules would hinder the proper execution of the algorithm's Gaussian sampling process, we set the number of fuzzy rules for each sub-classifier K by grid search from 10 to 30 with a step size of 2, and the number of sub-classifiers is set to $M = 3$. For the settings of the remaining parameters, we adopt the same settings as in [1].

For SVR, we adopt the RBF kernel as the kernel function, according to [43], we set its related parameter ψ by grid search from $\{0.001, 0.01, 0.1, 1, 10\}$. Additionally, we set ζ and ν by grid search from $\{0.01, 0.1, 1, 10, 100\}$ and $\{0.01, 0.05, 0.1, 0.5, 1\}$, respectively.

For KNN, the number of nearest neighbors k is set by grid search from $\{1, 3, 5, 7, 9, 11\}$.

In our experiments, for each of the utilized methods, five-fold cross-validation was performed on all adopted 20 datasets to obtain fair experimental results, where each fold was repeated five times to report average results and each feature of each dataset was normalized into $[0, 1]$. Additionally, the grid search strategy is adopted to achieve the best classification performance for each of the utilized methods.

4.1.3. Evaluation indices

To evaluate the classification performance of SP-TSK-RA, accuracy (ACC) and weighted F1-score are selected as the evaluation indices. The calculation method for the weighted F1-score is as follows:

$$\text{Weighted F1} = \sum_{i=1}^C \left(\frac{n_i}{N} \times \text{F1}_i \right) \quad (41)$$

$$\text{F1}_i = 2 \times \frac{P_i \times R_i}{P_i + R_i} \quad (42)$$

$$P_i = \frac{TP_i}{TP_i + FP_i}, \quad R_i = \frac{TP_i}{TP_i + FN_i} \quad (43)$$

where C and N represents the number of classes and samples, respectively. n_i represents the number of true samples in the i -th class. TP_i indicates the number of true positives for the i -th class, FP_i and FN_i represent the number of false positives and false negatives, respectively.

Additionally, the Friedman and Bonferroni-Dunn statistical tests [44] are adopted to conduct the statistical analysis of all adopted methods.

4.2. Results and discussions

4.2.1. On classification performance, statistical analysis, interpretability, and Identification of ambiguous samples

Within our experiments, to evaluate the classification performance of all utilized methods, we report the average training and testing accuracies (ACC) of all adopted methods on all adopted datasets in Table 2. Testing accuracy ranking values (Rank) of all adopted methods and the average numbers (R) of all adopted fuzzy rules are also reported in Table 4, where “–” means non-fuzzy rules are adopted. Additionally, to evaluate the capability of SP-TSK-RA in terms of generating robust antecedents on noisy datasets, examples on the Blood (BLO), Balance (BAL), and Banknote (BAN) datasets by introducing 10% and 20% Gaussian White Noise (GWN) are conducted. Besides, we introduce a new metric called Average Center Shift Distance (ACSD). Considering that the cluster indices generated in different runs are inherently unordered, we first employ the Hungarian algorithm [45] to find the optimal bipartite matching between the antecedent centers trained on the original dataset and those on the noise-injected datasets. The ACSD is then calculated by the average Euclidean distance of these optimally matched center pairs. Due to space limitations, the changes in the generated antecedent centers and the ACSD values between SP-TSK-RA and the two adopted comparative methods (i.e., First-TSK and B-TSK-FC) on the BLO dataset are reported in Table 5, while the results for the BAL and BAN datasets are provided in the supplementary file (i.e., Table S1 and Table S2). Furthermore, based on the BLO dataset with 20% GWN, we also investigate the distribution of the learned sample weights to validate the capability of the self-paced learning mechanism in identifying ambiguous samples.

Classification performance: From Table 2, Table 3, and Table 4, it is evident that SP-TSK-RA yields superior or at least competitive results compared with other rule-based comparative methods (i.e., L2-TSK, First-TSK, Zero-TSK, WL-TSK, and B-TSK-FC) in classification accuracy and weighted F1-score on most datasets. This advantage stems from SP-TSK-RA generating robust antecedents by fuzzy ensemble clustering and then refining consequent optimization by self-paced learning, which adaptively assesses sample reliability and learns progressively from easy to complex instances, thereby mitigating the impact of ambiguous samples. Additionally, it can be observed that Zero-TSK and First-TSK exhibit relatively inferior performance due to their lack of adaptations for ambiguous samples as classic TSK models. B-TSK-FC achieves comparable results to the previous two methods with fewer rules, due to its simultaneous optimization of antecedents and consequents, which enables the extraction of important rules. L2-TSK, due to the inclusion of a regularization term, substantially enhances the model's generalization capability over Zero-TSK and First-TSK. It is also noteworthy that WL-TSK outperforms most rule-based comparative methods in performance, primarily because the fuzzy-knowledge-out operation effectively prevents the co-adaptation of fuzzy rules, serving as a regularization technique that significantly improves the model's generalization capability and avoids overfitting. Moreover, SVR as a regression variant of SVM, outperforms all classifiers except SP-TSK-RA for most cases. Similarly, while the non-fuzzy baseline KNN achieves excellent results on some datasets, SP-TSK-RA still delivers superior performance across the majority of them. Besides, it is noteworthy that unlike the rule-based models, both SVR and KNN lack explicit linguistic interpretability.

Statistical analysis: Based on Table 2, we applied the Friedman and Bonferroni-Dunn statistical tests [44] to evaluate classification performance. According to the Friedman Test, if the p -value of the test is less than 0.05, it indicates that at least one algorithm performs significantly differently from the others. From Table 2 and [44], we get p -value < 0.005, it confirms the existence of significant differences among the methods. Furthermore, we employed the Bonferroni-Dunn test for a post-hoc analysis to compare the performance differences between SP-TSK-RA and the other seven methods. In this analysis, the significance threshold is set to 0.05, and the adjusted p -value for each comparative method is compared against this threshold to determine statistical significance. The adjusted p -values for L2-TSK, Zero-TSK, First-TSK, B-TSK-FC, WL-TSK, and KNN are 4.37×10^{-3} , 5.90×10^{-7} , 6.51×10^{-6} , 5.90×10^{-7} , 3.16×10^{-2} , and 3.86×10^{-2} , respectively. Since all adjusted p -values are less than 0.05, it confirms that there are significant differences between SP-TSK-RA and these six methods. Conversely, the adjusted p -value for SVR is 1, which exceeds the threshold, indicating no significant difference between SP-TSK-RA and SVR.

Interpretability: From Table 4 and Table 6, we demonstrate the interpretability of SP-TSK-RA by contrasting the average number of adopted fuzzy rules (R) alongside the model complexities (MC) across all adopted TSK-based algorithms. Table 4 presents the average number of fuzzy rules for each method across the 20 adopted datasets, while Table 6 reports their corresponding MC. Since SVR and

Table 2

Average training/testing ACCs (ACC) of all adopted methods on all 20 adopted datasets.

Datasets		BAN	BLO	DER	HEA	ILP	MAM	MAG	PAG	ECO	CON
L2-TSK	Training	0.9941 ±	0.7969 ±	0.7776 ±	0.8519 ±	0.7242 ±	0.8344 ±	0.8532 ±	0.9142 ±	0.5955 ±	0.4381 ±
	ACC	0.0013	0.0160	0.0166	0.0246	0.0085	0.0109	0.0014	0.0039	0.0190	0.0094
	Testing	0.9945 ±	0.7957 ±	0.7451 ±	0.8473 ±	0.7277 ±	0.7917 ±	0.8509 ±	0.9115 ±	0.5849 ±	0.4422 ±
	ACC	0.0050	0.0213	0.0262	0.0347	0.0181	0.0347	0.0039	0.0056	0.0331	0.0101
Zero-TSK	Training	0.9956 ±	0.8152 ±	0.6568 ±	0.8174 ±	0.7267 ±	0.8249 ±	0.8071 ±	0.8914 ±	0.6749 ±	0.2744 ±
	ACC	0.0033	0.0074	0.0387	0.0150	0.0143	0.0069	0.0023	0.0047	0.0411	0.0019
	Testing	0.9960 ±	0.7890 ±	0.6515 ±	0.8160 ±	0.7075 ±	0.8003 ±	0.8071 ±	0.8904 ±	0.6242 ±	0.2700 ±
	ACC	0.0024	0.0144	0.0463	0.0483	0.0090	0.0378	0.0068	0.0062	0.0470	0.0135
First-TSK	Training	0.9952 ±	0.8072 ±	0.7147 ±	0.8507 ±	0.7551 ±	0.8266 ±	0.8434 ±	0.9091 ±	0.6161 ±	0.2969 ±
	ACC	0.0005	0.0105	0.0642	0.0166	0.0165	0.0105	0.0018	0.0052	0.0202	0.0144
	Testing	0.9947 ±	0.7863 ±	0.6827 ±	0.8192 ±	0.7076 ±	0.8158 ±	0.8442 ±	0.9075 ±	0.6162 ±	0.2936 ±
	ACC	0.0057	0.0222	0.0663	0.0511	0.0399	0.0381	0.0054	0.0081	0.0418	0.0119
B-TSK-FC	Training	0.9989 ±	0.7605 ±	0.7412 ±	0.8568 ±	0.7091 ±	0.8233 ±	0.8484 ±	0.9168 ±	0.6795 ±	0.3189 ±
	ACC	0.0218	0.0041	0.0209	0.0156	0.0067	0.0130	0.0045	0.0107	0.0221	0.0103
	Testing	0.9903 ±	0.7674 ±	0.7119 ±	0.8437 ±	0.7183 ±	0.8223 ±	0.8478 ±	0.9139 ±	0.6627 ±	0.3037 ±
	ACC	0.0218	0.0035	0.0828	0.0808	0.0085	0.0432	0.0031	0.0198	0.0318	0.0154
SVR	Training	0.9973 ±	0.8022 ±	0.8197 ±	0.8722 ±	0.7616 ±	0.8302 ±	0.8585 ±	0.9205 ±	0.6957 ±	0.5873 ±
	ACC	0.0016	0.0226	0.0106	0.0675	0.0280	0.0131	0.0018	0.0041	0.0036	0.1431
	Testing	0.9964 ±	0.7928 ±	0.8058 ±	0.8444 ±	0.7307 ±	0.8250 ±	0.8543 ±	0.9183 ±	0.6816 ±	0.4868 ±
	ACC	0.0026	0.0129	0.0349	0.0483	0.0142	0.0373	0.0041	0.0029	0.0071	0.0133
WL-TSK	Training	0.9814 ±	0.7791 ±	0.7561 ±	0.8546 ±	0.7536 ±	0.8248 ±	0.8358 ±	0.9077 ±	0.6829 ±	0.4658 ±
	ACC	0.0014	0.0123	0.0184	0.0096	0.0124	0.0228	0.0029	0.0046	0.0296	0.0156
	Testing	0.9832 ±	0.7822 ±	0.7372 ±	0.8370 ±	0.7338 ±	0.8291 ±	0.8333 ±	0.9089 ±	0.6717 ±	0.4516 ±
	ACC	0.0087	0.0432	0.0750	0.0401	0.0501	0.0374	0.0091	0.0069	0.1081	0.0134
KNN	Training	0.9985 ±	0.8037 ±	0.8995 ±	0.8611 ±	0.7499 ±	0.8331 ±	0.8636 ±	0.9671 ±	0.8353 ±	0.6022 ±
	ACC	0.0007	0.0090	0.0059	0.0164	0.0143	0.0063	0.0015	0.0017	0.0108	0.0088
	Testing	0.9883 ±	0.7794 ±	0.8848 ±	0.8370 ±	0.6601 ±	0.8029 ±	0.8398 ±	0.9605 ±	0.8020 ±	0.5028 ±
	ACC	0.0028	0.0246	0.0197	0.0475	0.0294	0.0232	0.0041	0.0060	0.0340	0.0224
SP-TSK-RA	Training	0.9984 ±	0.8026 ±	0.8012 ±	0.8600 ±	0.7597 ±	0.8392 ±	0.8537 ±	0.9164 ±	0.7017 ±	0.5084 ±
	ACC	0.0036	0.0107	0.0084	0.0178	0.0168	0.0076	0.0038	0.0024	0.0295	0.0112
	Testing	0.9973 ±	0.8147 ±	0.7566 ±	0.8519 ±	0.7580 ±	0.8334 ±	0.8540 ±	0.9162 ±	0.6987 ±	0.4728 ±
	ACC	0.0061	0.0230	0.0720	0.0582	0.0296	0.0265	0.0063	0.0063	0.0423	0.0276
Datasets		TAE	V2C	AUT	BAL	V3C	WAV	FER	PLR	SST	ZOO
L2-TSK	Training	0.5922 ±	0.8317 ±	0.6780 ±	0.8149 ±	0.7536 ±	0.9248 ±	0.9025 ±	0.7140 ±	0.8909 ±	0.8977 ±
	ACC	0.0823	0.0331	0.0239	0.0063	0.0321	0.0040	0.0146	0.0044	0.0047	0.0655
	Testing	0.5049 ±	0.8285 ±	0.6737 ±	0.8195 ±	0.7496 ±	0.9158 ±	0.8982 ±	0.7124 ±	0.8887 ±	0.8325 ±
	ACC	0.0927	0.0360	0.0528	0.0151	0.0508	0.0006	0.0323	0.0165	0.0093	0.0672
Zero-TSK	Training	0.6495 ±	0.8469 ±	0.6991 ±	0.8305 ±	0.7677 ±	0.8710 ±	0.9026 ±	0.7257 ±	0.8986 ±	0.9417 ±
	ACC	0.1404	0.0463	0.0342	0.0131	0.0142	0.0081	0.0312	0.0230	0.0041	0.0563
	Testing	0.5245 ±	0.8388 ±	0.6725 ±	0.8215 ±	0.7373 ±	0.8678 ±	0.8813 ±	0.7057 ±	0.8942 ±	0.8285 ±
	ACC	0.0634	0.0228	0.0524	0.0176	0.0676	0.0098	0.0234	0.0240	0.0078	0.0580
First-TSK	Training	0.6073 ±	0.8760 ±	0.7127 ±	0.8251 ±	0.7921 ±	0.9075 ±	0.8938 ±	0.5973 ±	0.8962 ±	0.9635 ±
	ACC	0.1387	0.0108	0.0198	0.0073	0.0275	0.0018	0.0264	0.0302	0.0071	0.0322
	Testing	0.4735 ±	0.8477 ±	0.6713 ±	0.8187 ±	0.7580 ±	0.9031 ±	0.8890 ±	0.5625 ±	0.8939 ±	0.8282 ±
	ACC	0.0572	0.0322	0.0901	0.0147	0.0801	0.0100	0.0273	0.0895	0.0100	0.0459
B-TSK-FC	Training	0.4527 ±	0.8059 ±	0.6969 ±	0.8438 ±	0.7300 ±	0.8630 ±	0.8769 ±	0.7125 ±	0.8829 ±	0.8332 ±
	ACC	0.0191	0.0197	0.0251	0.0380	0.0081	0.0128	0.0000	0.0083	0.0060	0.0769

(continued on next page)

Table 2 (continued)

SVR	Testing	0.4650 ±	0.7984 ±	0.6800 ±	0.8148 ±	0.7241 ±	0.8707 ±	0.8750 ±	0.7062 ±	0.8796 ±	0.8156 ±
	ACC	0.0260	0.0489	0.0396	0.0291	0.0327	0.0183	0.0000	0.0232	0.0098	0.0617
	Training	0.5879 ±	0.8645 ±	0.7212 ±	0.8500 ±	0.7742 ±	0.9204 ±	0.9025 ±	0.7143 ±	0.9050 ±	0.8960 ±
	ACC	0.1420	0.0332	0.0279	0.0296	0.0086	0.0032	0.0479	0.0027	0.0033	0.0312
	Testing	0.5630 ±	0.8516 ±	0.6891 ±	0.8336 ±	0.7613 ±	0.9133 ±	0.8900 ±	0.7144 ±	0.9040 ±	0.8514 ±
	ACC	0.0786	0.0385	0.0301	0.0364	0.0417	0.0040	0.0418	0.0107	0.0082	0.0613
	Training	0.5759 ±	0.8508 ±	0.7056 ±	0.8196 ±	0.7121 ±	0.9001 ±	0.9075 ±	0.7280 ±	0.8926 ±	0.9039 ±
	ACC	0.0778	0.0208	0.0238	0.0064	0.0190	0.0062	0.0105	0.0209	0.0091	0.0295
	Testing	0.5596 ±	0.8516 ±	0.6960 ±	0.8224 ±	0.7194 ±	0.9017 ±	0.9010 ±	0.7191 ±	0.8922 ±	0.8604 ±
	ACC	0.0719	0.0289	0.0948	0.0510	0.0405	0.0091	0.0418	0.0866	0.0257	0.0150
	Training	0.6199 ±	0.8434 ±	0.8534 ±	0.8986 ±	0.8329 ±	0.9075 ±	0.8845 ±	0.6997 ±	0.9416 ±	0.8752 ±
	ACC	0.0238	0.0125	0.0157	0.0068	0.0145	0.0038	0.0092	0.0155	0.0020	0.0197
WL-TSK	Testing	0.5404 ±	0.8103 ±	0.7139 ±	0.8637 ±	0.7903 ±	0.8868 ±	0.8808 ±	0.6814 ±	0.9288 ±	0.8473 ±
	ACC	0.0735	0.0509	0.0372	0.0216	0.0479	0.0135	0.0198	0.0332	0.0047	0.0563
	Training	0.6250 ±	0.8730 ±	0.7383 ±	0.8392 ±	0.7490 ±	0.9408 ±	0.9197 ±	0.7192 ±	0.8919 ±	0.8538 ±
	ACC	0.0224	0.0097	0.0083	0.0196	0.0291	0.0067	0.0132	0.0147	0.0132	0.0342
SP-TSK-RA	Testing	0.6153 ±	0.8670 ±	0.7188 ±	0.8287 ±	0.7416 ±	0.9238 ±	0.9140 ±	0.7262 ±	0.8801 ±	0.8537 ±
	ACC	0.0391	0.0263	0.0574	0.0195	0.0913	0.0110	0.0323	0.0403	0.0127	0.0551

Table 3

Average F1-scores of all adopted methods on all 20 adopted datasets.

Datasets		BAN	BLO	DER	HEA	ILP	MAM	MAG	PAG	ECO	CON
L2-TSK	Training	0.9941 ± 0.0013	0.7607 ± 0.0302	0.7774 ± 0.0166	0.8490 ± 0.0267	0.6352 ± 0.0330	0.8345 ± 0.0108	0.8490 ± 0.0014	0.8957 ± 0.0080	0.6233 ± 0.0662	0.3191 ± 0.0179
	F1-scores										
Zero-TSK	Testing	0.9945 ± 0.0050	0.7608 ± 0.0327	0.7429 ± 0.0788	0.8431 ± 0.0295	0.6420 ± 0.0375	0.7918 ± 0.0346	0.8466 ± 0.0043	0.8929 ± 0.0084	0.6065 ± 0.0696	0.3258 ± 0.0104
	F1-scores										
First-TSK	Training	0.9956 ± 0.0033	0.7962 ± 0.0107	0.6302 ± 0.0469	0.8161 ± 0.0148	0.6800 ± 0.0255	0.8251 ± 0.0069	0.7981 ± 0.0027	0.8962 ± 0.0049	0.7324 ± 0.0400	0.1822 ± 0.0047
	F1-scores										
B-TSK-FC	Testing	0.9960 ± 0.0024	0.7638 ± 0.0173	0.6172 ± 0.0558	0.8143 ± 0.0488	0.6525 ± 0.0154	0.8003 ± 0.0377	0.7981 ± 0.0070	0.8945 ± 0.0039	0.6673 ± 0.0532	0.1723 ± 0.0219
	F1-scores										
SVR	Training	0.9952 ± 0.0005	0.7783 ± 0.0189	0.6996 ± 0.0643	0.8503 ± 0.0165	0.7217 ± 0.0204	0.8266 ± 0.0106	0.8372 ± 0.0019	0.9129 ± 0.0032	0.6755 ± 0.0187	0.2384 ± 0.0230
	F1-scores										
WL-TSK	Testing	0.9947 ± 0.0057	0.7537 ± 0.0227	0.6568 ± 0.0716	0.8164 ± 0.0532	0.6592 ± 0.0618	0.8156 ± 0.0384	0.8381 ± 0.0056	0.9103 ± 0.0067	0.6634 ± 0.0446	0.2327 ± 0.0093
	F1-scores										
KNN	Training	0.9989 ± 0.0218	0.6741 ± 0.0255	0.7406 ± 0.0217	0.8535 ± 0.0151	0.6090 ± 0.0219	0.8226 ± 0.0127	0.8393 ± 0.0035	0.9127 ± 0.0098	0.6646 ± 0.0239	0.2468 ± 0.0185
	F1-scores										
SP-TSK-RA	Testing	0.9901 ± 0.0217	0.6784 ± 0.0198	0.7052 ± 0.0909	0.8423 ± 0.0821	0.6207 ± 0.0381	0.8222 ± 0.0434	0.8388 ± 0.0044	0.9106 ± 0.0163	0.6509 ± 0.0328	0.2306 ± 0.0185
	F1-scores										
Zero-TSK	Training	0.9973 ± 0.0016	0.7648 ± 0.0413	0.8154 ± 0.0113	0.8715 ± 0.0678	0.7053 ± 0.0630	0.8302 ± 0.0131	0.8536 ± 0.0020	0.9007 ± 0.0180	0.6026 ± 0.0049	0.5430 ± 0.1861
	F1-scores										
First-TSK	Testing	0.9964 ± 0.0026	0.7563 ± 0.0126	0.7939 ± 0.0384	0.8432 ± 0.0492	0.6703 ± 0.0424	0.8150 ± 0.0371	0.8491 ± 0.0038	0.8987 ± 0.0165	0.5770 ± 0.0143	0.4389 ± 0.0526
	F1-scores										
B-TSK-FC	Training	0.9814 ± 0.0014	0.7191 ± 0.0214	0.7437 ± 0.0152	0.8537 ± 0.0094	0.6443 ± 0.0163	0.8249 ± 0.0228	0.8292 ± 0.0032	0.8976 ± 0.0036	0.6782 ± 0.0252	0.4432 ± 0.0259
	F1-scores										
KNN	Testing	0.9833 ± 0.0087	0.7253 ± 0.0564	0.7284 ± 0.0622	0.8360 ± 0.0412	0.6254 ± 0.0667	0.8232 ± 0.0374	0.8264 ± 0.0103	0.8983 ± 0.0063	0.6730 ± 0.0960	0.4330 ± 0.0138
	F1-scores										
SP-TSK-RA	Training	0.9985 ± 0.0007	0.7820 ± 0.0115	0.9099 ± 0.0058	0.8606 ± 0.0165	0.7318 ± 0.0165	0.8326 ± 0.0062	0.8580 ± 0.0016	0.9653 ± 0.0020	0.8291 ± 0.0114	0.6005 ± 0.0093
	F1-scores										
Zero-TSK	Testing	0.9883 ± 0.0028	0.7542 ± 0.0275	0.8853 ± 0.0191	0.8363 ± 0.0479	0.6366 ± 0.0335	0.8024 ± 0.0235	0.8325 ± 0.0044	0.9575 ± 0.0068	0.8029 ± 0.0376	0.5001 ± 0.0228
	F1-scores										
First-TSK	Training	0.9984 ± 0.0036	0.7808 ± 0.0207	0.8009 ± 0.0077	0.8591 ± 0.0171	0.7382 ± 0.0232	0.8362 ± 0.0077	0.8510 ± 0.0037	0.8908 ± 0.0045	0.6830 ± 0.0556	0.4970 ± 0.0084
	F1-scores										
B-TSK-FC	Testing	0.9973 ± 0.0061	0.7924 ± 0.0306	0.7530 ± 0.0772	0.8501 ± 0.0584	0.7352 ± 0.0348	0.8230 ± 0.0268	0.8512 ± 0.0068	0.8901 ± 0.0111	0.6782 ± 0.0675	0.4636 ± 0.0389
	F1-scores										
Datasets		TAE	V2C	AUT	BAL	V3C	WAV	FER	PLR	SST	ZOO
L2-TSK	Training	0.5947 ± 0.0850	0.8258 ± 0.0417	0.6790 ± 0.0362	0.7838 ± 0.0062	0.7484 ± 0.0434	0.9253 ± 0.0040	0.8732 ± 0.0304	0.5993 ± 0.0136	0.9110 ± 0.0042	0.8894 ± 0.0767
	F1-scores										
Zero-TSK	Testing	0.4889 ± 0.1036	0.8227 ± 0.0426	0.6742 ± 0.0375	0.7878 ± 0.0147	0.7456 ± 0.0607	0.9163 ± 0.0005	0.8576 ± 0.0569	0.5929 ± 0.0216	0.9097 ± 0.0044	0.8149 ± 0.0857
	F1-scores										
First-TSK	Training	0.6554 ± 0.1417	0.8468 ± 0.0459	0.7080 ± 0.0419	0.7993 ± 0.0130	0.7685 ± 0.0145	0.8736 ± 0.0078	0.8702 ± 0.0578	0.6451 ± 0.0543	0.9177 ± 0.0039	0.9397 ± 0.0574
	F1-scores										
B-TSK-FC	Testing	0.5137 ± 0.0661	0.8379 ± 0.0270	0.6758 ± 0.0383	0.7889 ± 0.0177	0.7351 ± 0.0752	0.8705 ± 0.0094	0.8431 ± 0.0426	0.6220 ± 0.0507	0.9137 ± 0.0032	0.8224 ± 0.0667
	F1-scores										
Zero-TSK	Training	0.5987 ± 0.1688	0.8760 ± 0.0113	0.7296 ± 0.0187	0.8031 ± 0.0044	0.7928 ± 0.0272	0.9086 ± 0.0018	0.8863 ± 0.0228	0.5930 ± 0.0172	0.9163 ± 0.0087	0.9631 ± 0.0331
	F1-scores										
First-TSK	Testing	0.4423 ± 0.0879	0.8470 ± 0.0349	0.6901 ± 0.0722	0.7967 ± 0.0103	0.7562 ± 0.0846	0.9043 ± 0.0095	0.8527 ± 0.0486	0.5444 ± 0.0743	0.9133 ± 0.0113	0.8012 ± 0.0464
	F1-scores										
B-TSK-FC	Training	0.4219 ± 0.0326	0.8014 ± 0.0214	0.6965 ± 0.0178	0.8083 ± 0.0395	0.7128 ± 0.0117	0.8652 ± 0.0122	0.8194 ± 0.0000	0.5966 ± 0.0051	0.9044 ± 0.0047	0.8218 ± 0.0840
	F1-scores										

(continued on next page)

Table 3 (continued)

SVR	Testing F1-scores	0.4188 ± 0.0490	0.7905 ± 0.0531	0.6875 ± 0.0341	0.7767 ± 0.0317	0.6976 ± 0.0345	0.8730 ± 0.0181	0.8167 ± 0.0000	0.5865 ± 0.0336	0.9024 ± 0.0042	0.7996 ± 0.0761
	Training F1-scores	0.5679 ± 0.1704	0.8610 ± 0.0380	0.6865 ± 0.0597	0.8374 ± 0.0205	0.7412 ± 0.0094	0.9211 ± 0.0029	0.8580 ± 0.0728	0.5953 ± 0.0036	0.8813 ± 0.0047	0.8629 ± 0.0371
	Testing F1-scores	0.5211 ± 0.1066	0.8502 ± 0.0388	0.6488 ± 0.0514	0.8215 ± 0.0283	0.7201 ± 0.0530	0.9140 ± 0.0037	0.8443 ± 0.0711	0.5954 ± 0.0141	0.8799 ± 0.0112	0.8048 ± 0.0724
WL-TSK	Training F1-scores	0.5521 ± 0.1167	0.8460 ± 0.0255	0.7018 ± 0.0297	0.7888 ± 0.0067	0.7045 ± 0.0190	0.9013 ± 0.0060	0.8611 ± 0.0194	0.6283 ± 0.0328	0.9068 ± 0.0093	0.9021 ± 0.0265
	Testing F1-scores	0.5253 ± 0.0912	0.8472 ± 0.0317	0.6955 ± 0.1143	0.7911 ± 0.0498	0.7061 ± 0.0574	0.9026 ± 0.0088	0.8436 ± 0.0628	0.6163 ± 0.1272	0.9058 ± 0.0348	0.8468 ± 0.0251
KNN	Training F1-scores	0.6122 ± 0.0249	0.8437 ± 0.0125	0.8543 ± 0.0150	0.8664 ± 0.0056	0.8337 ± 0.0143	0.9082 ± 0.0038	0.8337 ± 0.0182	0.6095 ± 0.0230	0.9462 ± 0.0027	0.8583 ± 0.0235
	Testing F1-scores	0.5206 ± 0.0749	0.8109 ± 0.0507	0.7094 ± 0.0396	0.8349 ± 0.0205	0.7906 ± 0.0489	0.8877 ± 0.0133	0.8250 ± 0.0283	0.5836 ± 0.0248	0.9214 ± 0.0068	0.8279 ± 0.0614
SP-TSK-RA	Training F1-scores	0.6230 ± 0.0228	0.8724 ± 0.0103	0.7522 ± 0.0070	0.8119 ± 0.0147	0.7546 ± 0.0291	0.9415 ± 0.0063	0.9013 ± 0.0203	0.6295 ± 0.0368	0.8981 ± 0.0135	0.8459 ± 0.0339
	Testing F1-scores	0.6066 ± 0.0381	0.8650 ± 0.0302	0.7387 ± 0.0495	0.7963 ± 0.0155	0.7433 ± 0.0963	0.9216 ± 0.0057	0.8857 ± 0.0583	0.6364 ± 0.0690	0.8927 ± 0.0146	0.8266 ± 0.0828

Table 4

Average fuzzy rules (R), and Testing Accuracy ranking values (Rank) of all adopted methods on all 20 adopted datasets.

Datasets	L2-TSK		Zero TSK		First TSK		B-TSK-FC		SVR		WL-TSK		KNN		SP-TSK-RA	
	R	Rank	R	Rank	R	Rank	R	Rank	R	Rank	R	Rank	R	Rank	R	Rank
BAN	8.2	5	32.2	3	16	4	14.4	6	--	2	15	8	--	7	9.6	1
BLO	13	2	32.4	4	12.4	5	5.8	8	--	3	13.6	6	--	7	5.2	1
DER	12	4	27	8	27.2	7	10.8	6	--	2	19.5	5	--	1	6.6	3
HEA	8.2	2	11.6	8	5.8	7	7.4	4	--	3	39.4	5	--	6	3.6	1
ILP	7.2	4	19.8	7	11.6	6	4.4	5	--	3	12.4	2	--	8	6.2	1
MAM	14.6	8	22.4	7	16.8	5	6.6	4	--	3	17.2	2	--	6	7	1
MAG	16.2	3	38.6	8	13.6	5	9.4	4	--	1	30.2	7	--	6	13.2	2
PAG	31.6	5	47.6	8	25.4	7	16	4	--	2	19.6	6	--	1	9.6	3
ECO	17.6	8	32.2	6	23.2	7	12.2	5	--	3	29.8	4	--	1	9	2
CON	14.8	5	46.6	8	24.6	7	10.2	6	--	2	32	4	--	1	6.5	3
TAE	35	6	42.6	5	34.2	7	24.4	8	--	2	21.2	3	--	4	15	1
V2C	8.8	6	24.6	5	8.8	4	8.6	8	--	2	26.2	3	--	7	4.8	1
AUT	30.2	6	34.2	7	30.2	8	17.2	5	--	4	14.6	3	--	2	11.6	1
BAL	18.8	6	27.4	5	21.4	7	9.8	8	--	2	23	4	--	1	7.6	3
V3C	13.2	4	15	6	10.6	3	7	7	--	2	25.2	8	--	1	8.8	5
WAV	9.4	2	35.2	8	9.8	4	9.2	7	--	3	24	5	--	6	6.9	1
FER	6	3	12.6	6	4	5	2	8	--	4	13.2	2	--	7	3.4	1
PLR	5.6	4	15.4	6	9.4	8	2.6	5	--	3	22	2	--	7	4	1
SST	38.2	6	26.2	3	16.4	4	20.4	8	--	2	33.6	5	--	1	19.4	7
ZOO	14.8	5	23.4	6	22.4	7	15.8	8	--	3	25	1	--	4	9	2
Average Rules/Ranks	16.17	4.7	28.35	6.2	17.19	5.85	10.71	6.2	--	2.55	22.83	4.25	--	4.2	8.35	2.05

Note: "--" indicates the corresponding method adopts no fuzzy rules.

Table 5

Antecedent centers(C) and Average Center Shift Distance (ACSD) for adopted methods on the BLO dataset under different GWN levels.

	First-TSK		B-TSK-FC		SP-TSK-RA	
	Antecedent centers	ACSD	Antecedent centers	ACSD	Antecedent centers	ACSD
Blo	C1:0.1780,0.1129,0.1129,0.7397 C2:0.0447,0.0747,0.0747,0.1588 C3:0.0943,0.3142,0.3142,0.7232 C4:0.1873,0.0776,0.0776,0.3605 C5:0.1720,0.0946,0.0946,0.5936 C6:0.1851,0.0100,0.0100,0.1333 C7:0.0499,0.1174,0.1174,0.3246 C8:0.2917,0.0113,0.0113,0.2237 C9:0.0749,0.1400,0.1400,0.5051 C10:0.0386,0.8787,0.8787,0.9924 C11:0.0678,0.2458,0.2458,0.8946 C12:0.0415,0.0072,0.0072,0.0181	--	C1:0.2404,0.0207,0.0207,0.2062 C2:0.0850,0.3640,0.3640,0.8277 C3:0.1578,0.1114,0.1114,0.6724 C4:0.0505,0.0224,0.0224,0.0583 C5:0.0810,0.1196,0.1196,0.3600	--	C1:0.1931,0.0723,0.0723,0.4182 C2:0.1142,0.1173,0.1173,0.3599 C3:0.0849,0.1631,0.1631,0.4741 C4:0.0734,0.1570,0.1570,0.4211 C5:0.1461,0.0730,0.0730,0.3873	--
Blo(10% GWN)	C1:0.2062,0.1234,0.1276,0.7349 C2:0.0561,0.0900,0.0958,0.1945 C3:0.1334,0.3230,0.3281,0.7158 C4:0.1950,0.1019,0.1023,0.3844 C5:0.1438,0.1396,0.1462,0.6017 C6:0.1899,0.0349,0.0419,0.1761 C7:0.0730,0.1488,0.1521,0.3304 C8:0.2965,0.0369,0.0408,0.2549 C9:0.0530,0.1364,0.1411,0.4895 C10:0.0523,0.8642,0.8662,0.9645 C11:0.1046,0.2571,0.2629,0.8773 C12:0.0525,0.0315,0.0371,0.0664	0.0478	C1:0.2013,0.0442,0.0499,0.2487 C2:0.1201,0.3774,0.3825,0.7862 C3:0.1705,0.1313,0.1368,0.6834 C4:0.0647,0.0444,0.0500,0.1016 C5:0.0569,0.1415,0.1454,0.3855	0.0541	C1:0.1998,0.0830,0.0787,0.4222 C2:0.1172,0.1171,0.1219,0.3504 C3:0.0922,0.1624,0.1766,0.4840 C4:0.0783,0.1553,0.1803,0.4252 C5:0.1401,0.0785,0.0992,0.3912	0.0192
Blo(20% GWN)	C1:0.1215,0.3492,0.3312,0.7288 C2:0.0619,0.1230,0.1247,0.2352 C3:0.1341,0.1637,0.1630,0.5409 C4:0.1304,0.2282,0.2188,0.5082 C5:0.2260,0.1402,0.1503,0.7221 C6:0.1999,0.0700,0.0781,0.1842 C7:0.2015,0.1272,0.1333,0.4020 C8:0.3043,0.0714,0.0742,0.2977 C9:0.1698,0.1570,0.1547,0.3575 C10:0.0757,0.9144,0.8641,0.9440 C11:0.0898,0.2920,0.2719,0.8643 C12:0.0878,0.0665,0.0759,0.1007	0.0983	C1:0.2624,0.0839,0.0853,0.2869 C2:0.0703,0.4417,0.3900,0.9193 C3:0.1136,0.2029,0.1428,0.7744 C4:0.1200,0.0828,0.0900,0.1424 C5:0.1312,0.1715,0.1708,0.4606	0.1340	C1:0.1956,0.0699,0.0662,0.4356 C2:0.1218,0.1291,0.1460,0.3615 C3:0.0804,0.1513,0.2055,0.5088 C4:0.0816,0.1151,0.1646,0.4182 C5:0.1608,0.0777,0.0812,0.3803	0.0338

KNN are not rule-based models, they lack linguistic interpretability and are thus excluded from these comparisons. According to Table 4 and Table 6, we can observe that SP-TSK-RA uses fewer rules than L2-TSK, Zero-TSK, B-TSK-FC, WL-TSK, and First-TSK in most cases. Consequently, SP-TSK-RA's model complexity is lower than most rule-based models, except for being slightly higher than B-TSK-FC. Even on the datasets where its fuzzy rules are not fewer or its model complexity is not the lowest, SP-TSK-RA still achieves better classification performance.

Next, to demonstrate the linguistic interpretability and the transparent reasoning process of SP-TSK-RA, we provide a concrete example on the V2C dataset. The features (F1 to F6) are normalized to [0, 1] and mapped to five fuzzy sets, where the center ranges of these five fuzzy sets are defined as [0, 0.2], (0.2, 0.4], (0.4, 0.6], (0.6, 0.8], and (0.8, 1], respectively, and their fuzzy sets have the corresponding linguistic meanings, i.e., Very Low (VL), Low (L), Medium (M), High (H), and Very High (VH). Therefore, Table 7 reports the four major fuzzy rules with high weights of SP-TSK-RA on V2C dataset. From Table 7, we take Rule 1 as an example and its linguistic meaning can be interpreted by as follows:

IF (F1 is Very Low) AND (F2 is Low) AND (F3 is Low) AND (F4 is Low) AND (F5 is Medium) AND (F6 is Very Low), THEN $f_1(\mathbf{x})$.

Clearly, this kind of fuzzy rule is straightforward and transparent to analyze and comprehend. Then, to further illustrate how SP-TSK-RA makes decisions, we take the following two test samples to detail the prediction paths:

$$\mathbf{x}_1 = [0.2645, 0.4825, 0.1709, 0.1827, 0.4397, 0.0421]$$

$$\mathbf{x}_2 = [0.3561, 0.5531, 0.3580, 0.2340, 0.3908, 0.0625]$$

As for test sample $\mathbf{x}_1$, based on its firing levels $\tilde{\mu}_k(\mathbf{x}_1)$ and rule weight in each fuzzy rule, its predicted value can be calculated as: $\hat{y}(\mathbf{x}_1) = 1 \cdot \tilde{\mu}_1(\mathbf{x}_1)f_1(\mathbf{x}_1) + 1 \cdot \tilde{\mu}_2(\mathbf{x}_1)f_2(\mathbf{x}_1) + 1 \cdot \tilde{\mu}_3(\mathbf{x}_1)f_3(\mathbf{x}_1) + 0.98 \cdot \tilde{\mu}_4(\mathbf{x}_1)f_4(\mathbf{x}_1) = -0.8205$. In accordance with the introduced minimal distance-based strategy, the category of $\mathbf{x}_1$ will be classified as -1. For test sample $\mathbf{x}_2$, its predicted value can be calculated in the same way as -1.2203, so the label of $\mathbf{x}_2$ will be predicted as -1.

Additionally, Table 5 illustrates the antecedent centers and their corresponding ACSD values obtained by the adopted methods on the BLO dataset under different noise levels. SP-TSK-RA maintains stable antecedent centers across different noise levels. Remarkably, even with the introduction of 20% GWN, most of the centers obtained by SP-TSK-RA exhibit no significant deviation. Similar stable

Table 6

Average running times (Time), and model complexities (MC) of all adopted methods on all 20 adopted datasets.

Data -sets	L2-TSK		Zero-TSK		First-TSK		B-TSK-FC		WL-TSK		SVR		KNN		SP-TSK-RA	
	Time	MC	Time	MC	Time	MC	Time	MC	Time	MC	Time	MC	Time	MC	Time	MC
BAN	0.034	106.6	0.082	289.8	0.070	208	1.496	129.6	0.053	135	0.810	--	0.005	--	0.260	124.8
BLO	0.024	169	0.035	291.6	0.027	161.2	0.370	52.2	0.031	122.4	0.178	--	0.004	--	0.080	67.6
DER	0.022	1236	0.030	1863	0.041	2801.6	0.533	745.2	0.064	1345.5	0.023	--	0.003	--	0.112	679.8
HEA	0.009	328	0.013	313.2	0.008	232	0.184	199.8	0.031	1063.8	0.013	--	0.003	--	0.043	144
ILP	0.019	223.2	0.037	415.8	0.028	359.6	0.313	92.4	0.044	260.4	0.081	--	0.004	--	0.094	192.2
MAM	0.035	233.6	0.034	246.4	0.037	268.8	0.891	72.6	0.048	189.2	0.128	--	0.005	--	0.101	112
MAG	7.086	502.2	1.835	810.6	6.788	421.6	12.244	197.4	0.993	634.2	4.562	--	0.105	--	6.842	409.2
PAG	1.478	979.6	0.680	999.6	1.363	787.4	3.227	336	0.093	411.6	2.068	--	0.015	--	3.375	297.6
ECO	0.012	387.2	0.019	483	0.016	510.4	0.212	183	0.001	447	0.050	--	0.003	--	0.136	198
CON	0.097	414.4	0.167	885.4	0.197	688.8	0.999	193.8	0.075	608	0.699	--	0.006	--	0.426	182
TAE	0.012	560	0.010	468.6	0.007	547.2	0.260	268.4	0.001	233.2	0.353	--	0.003	--	0.118	240
V2C	0.008	167.2	0.015	319.8	0.008	167.2	0.392	111.8	0.009	340.6	0.022	--	0.003	--	0.035	91.2
AUT	0.027	664.4	0.026	513	0.028	664.4	0.364	258	0.057	219	0.014	--	0.003	--	0.158	255.2
BAL	0.037	244.4	0.038	246.6	0.071	278.2	0.506	88.2	0.043	207	0.009	--	0.004	--	0.061	98.8
V3C	0.008	250.8	0.010	195	0.007	201.4	0.378	91	0.003	327.6	0.012	--	0.003	--	0.096	167.2
WAV	0.205	601.6	0.320	1513.6	0.226	627.2	2.357	395.6	0.131	1032	1.130	--	0.010	--	1.200	441.6
FER	0.004	168	0.007	239.4	0.005	112	0.049	38	0.003	250.8	0.022	--	0.003	--	0.028	95.2
PLR	0.005	207.2	0.006	385	0.007	347.8	0.101	65	0.034	550	0.012	--	0.003	--	0.082	148
SST	0.324	1069.6	0.119	497.8	0.137	459.2	3.257	387.6	0.081	638.4	0.203	--	0.007	--	2.308	543.2
ZOO	0.008	725.2	0.008	772.2	0.010	1097.6	0.312	521.4	0.003	825	0.021	--	0.002	--	0.154	441

Table 7
Major Fuzzy Rules of SP-TSK-RA on V2C.

IF-parts								Then-parts
Rule	Rule weight	F1	F2	F3	F4	F5	F6	
1	1	VL	L	L	L	M	VL	$f_1(x) = -2.0965 - 0.5937F1 - 3.0187F2 - 1.5505F3 + 0.6173F4 + 7.0429F5 - 6.6795F6$
2	1	M	H	M	L	M	VL	$f_2(x) = 1.3468 - 0.7257F1 - 0.5370F2 - 2.1487F3 - 0.1801F4 + 0.5086F5 - 5.6905F6$
3	1	L	M	L	L	H	VL	$f_3(x) = 0.0306 + 0.7414F1 - 0.9154F2 + 0.1899F3 + 1.1876F4 - 0.0197F5 - 4.0686F6$
4	0.98	L	M	L	VL	M	VL	$f_4(x) = -0.6616 + 0.6690F1 + 0.1508F2 + 0.5895F3 + 0.4425F4 + 0.5182F5 - 2.3729F6$

performance is also consistently observed on the BAL and BAN datasets in the [supplementary material](#). This stability is attributed to the fuzzy ensemble clustering mechanism employed in the antecedent generation phase, which mitigates the negative impacts of noise by integrating outputs from multiple single clustering algorithms. In contrast, the antecedent centers of the other two comparative methods show noticeable shifts due to noise perturbations. From [Table 4](#) and [Table 6](#), it can be concluded that SP-TSK-RA not only generates stable and interpretable antecedents but also utilizes fewer rules with lower model complexity, thereby demonstrating superior interpretability among all the adopted methods.

Identification of Ambiguous Samples: [Fig. 2](#) presents the distribution of the final weights of all training samples across different intervals on the original BLO dataset and the BLO dataset with 20% GWN. From [Fig. 2\(a\)](#), the majority of samples are assigned high weights, indicating that most samples are reliable and easy to classify. However, from [Fig. 2\(b\)](#), when 20% GWN is introduced, the weight distribution shifts and the number of high-weight samples decrease, while the number of samples in the lower-weight bins increases. This indicates that the proposed SPL mechanism in SP-TSK-RA effectively identifies the injected ambiguous samples (noisy samples) and adaptively suppresses their negative impacts by assigning them lower weights during the training process, which also confirms the effectiveness of the SPL mechanism.

4.2.2. On running time

During our experiments, to assess computational efficiency, we averaged the execution times of all comparative algorithms over five trials across the 20 datasets, with the detailed outcomes presented in [Table 6](#).

From [Table 6](#), it can be observed that KNN exhibits the fastest runtime among all adopted methods since it is a lazy learning algorithm that lacks an explicit model parameter optimization process. Among the rule-based methods, L2-TSK, Zero-TSK, and First-TSK exhibit faster runtimes among the adopted methods on most datasets. This is because these methods employ single clustering algorithms for antecedent generation and simple optimization techniques for consequents. WL-TSK also exhibits efficient runtime performance, particularly on large-scale datasets, attributable to WL-TSK utilizing fixed fuzzy partitions, transforming parameter learning into a linear problem that can be solved directly via efficient matrix operations. SVR exhibits longer runtime compared to the previous five methods, because it involves solving a quadratic programming problem with kernel matrix computations. Additionally, it can be observed that B-TSK-FC has the longest runtime, because it employs MCMC sampling for global optimization and performs multiple MH sampling steps [\[40\]](#), significantly increasing computational cost. SP-TSK-RA exhibits faster runtime than B-TSK-FC but slower than the other methods, because its antecedent construction employs a fuzzy ensemble clustering algorithm that integrates results from multiple clustering techniques. However, it is worth noting that this increased computational cost of SP-TSK-RA is primarily incurred during the training process, while it still achieves a faster test speed due to fewer major fuzzy rules. To empirically support the above claim, due to space limitations, we separately recorded the training and test times on several datasets with varying sample sizes and feature sizes (i.e., V2C, BAN, WAV, and MAG) in [Fig. S1](#) of the [supplementary file](#). From [Fig. S1](#) of the [supplementary file](#), the testing times are remarkably shorter than the training times, which confirms that once trained, the test phase of SP-TSK-RA is fast and highly efficient. SP-TSK-RA yields a significantly more compact rule base and more stable antecedent centers, thereby ensuring high structural simplicity and interpretability for the final model. Overall, although SP-TSK-RA's runtime is longer, it remains acceptable. Considering factors such as classification performance and interpretability, the time investment is worthwhile.

Based on the above analysis, we can conclude that SP-TSK-RA achieves at least comparable classification performance to other methods while maintaining superior interpretability and acceptable runtime efficiency.

4.2.3. Ablation study

While both fuzzy ensemble clustering and self-paced learning play important roles in SP-TSK-RA, this subsection is organized to further evaluate the effectiveness of these two components. We construct two ablation variants of the proposed method. For the first variant, named SP-TSK-RA1, the fuzzy ensemble clustering mechanism is replaced by a single clustering algorithm. For the second variant, named SP-TSK-RA2, the self-paced learning mechanism is removed from the consequent parameter learning framework. Five-fold cross-validation is adopted to report the experimental results of the above three methods on all 20 adopted datasets in [Table 8](#).

From [Table 8](#), it can be observed that SP-TSK-RA outperforms both SP-TSK-RA1 and SP-TSK-RA2 in most cases. These results demonstrate that both the fuzzy ensemble clustering strategy and the self-paced learning mechanism are effective and essential components for enhancing the overall performance.

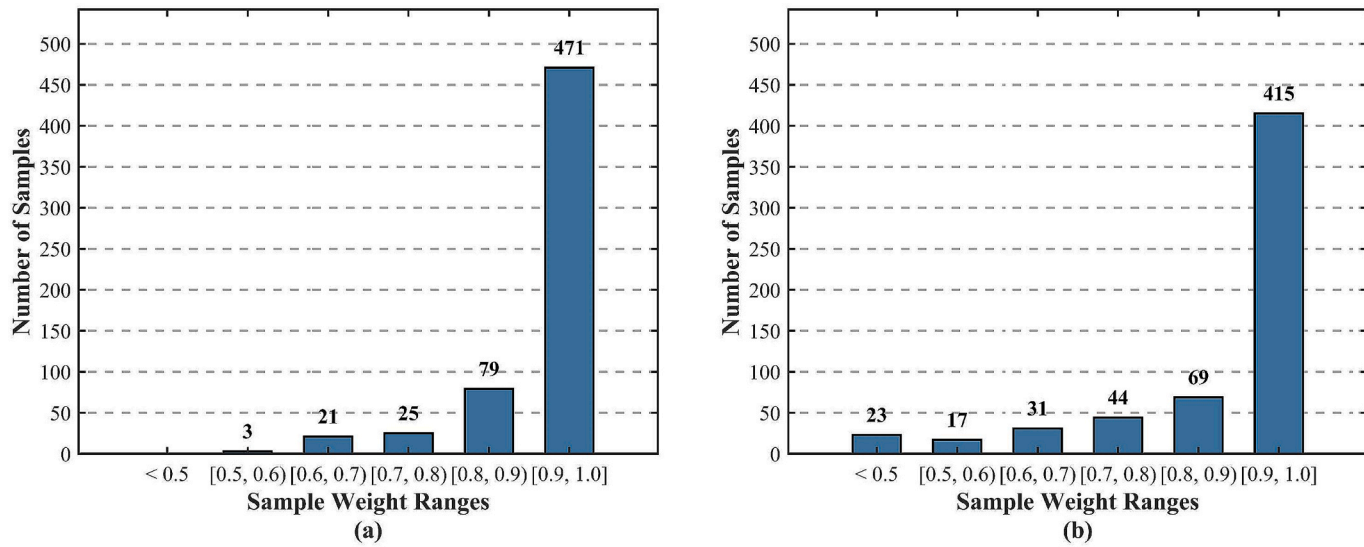


Fig. 2. The distributions of sample weights on the training set of BLO dataset under different conditions. (a) The original dataset. (b) The dataset with 20% Gaussian White Noise (GWN).

Table 8
Average training/testing ACCs (ACC) of SP-TSK-RA, SP-TSK-RA1, and SP-TSK-RA2 on all 20 adopted datasets.

Datasets		<i>BAN</i>	<i>BLO</i>	<i>DER</i>	<i>HEA</i>	<i>ILP</i>	<i>MAM</i>	<i>MAG</i>	<i>PAG</i>	<i>ECO</i>	<i>CON</i>
SP-TSK-RA	Training	0.9984 ±	0.8026 ±	0.8012 ±	0.8600 ±	0.7597 ±	0.8392 ±	0.8537 ±	0.9164 ±	0.7017 ±	0.5084 ±
	ACC	0.0036	0.0107	0.0084	0.0178	0.0168	0.0076	0.0038	0.0024	0.0295	0.0112
	Testing	0.9973 ±	0.8147 ±	0.7566 ±	0.8519 ±	0.7580 ±	0.8334 ±	0.8540 ±	0.9162 ±	0.6987 ±	0.4728 ±
SP-TSK-RA1	ACC	0.0061	0.0230	0.0720	0.0582	0.0296	0.0265	0.0063	0.0063	0.0423	0.0276
	Training	0.9966	0.8023	0.7622	0.8232	0.7406	0.8149 ±	0.8528	0.9076	0.6162	0.4705
	ACC	±0.0019	±0.0051	±0.0214	±0.0127	±0.0162	0.0095	±0.0018	±0.0029	±0.0227	±0.0103
SP-TSK-RA2	Testing	0.9965	0.7807	0.7417	0.8519	0.7357	0.8130 ±	0.8623	0.9060	0.6542	0.4300
	ACC	±0.0015	±0.0190	±0.0740	±0.0473	±0.0156	0.0371	±0.0042	±0.0056	±0.0268	±0.0174
	Training	0.9920	0.7818	0.7208	0.8623	0.7159	0.8994 ±	0.8533	0.9034	0.5799	0.4075
SP-TSK-RA2	ACC	±0.0080	±0.0072	±0.0177	±0.0118	±0.0141	0.0275	±0.0034	±0.0037	±0.0282	±0.0113
	Testing	0.9891	0.7530	0.6895	0.8290	0.6694	0.7787 ±	0.8531	0.9076	0.5759	0.3831
	ACC	±0.0110	±0.0268	±0.0890	±0.0789	±0.0321	0.0874	±0.0062	±0.0042	±0.0550	±0.0450
Datasets		<i>TAE</i>	<i>V2C</i>	<i>AUT</i>	<i>BAL</i>	<i>V3C</i>	<i>WAV</i>	<i>FER</i>	<i>PLR</i>	<i>SST</i>	<i>ZOO</i>
SP-TSK-RA	Training	0.6250 ±	0.8730 ±	0.7383 ±	0.8392 ±	0.7490 ±	0.9408 ±	0.9197 ±	0.7192 ±	0.8919 ±	0.8538 ±
	ACC	0.0224	0.0097	0.0083	0.0196	0.0291	0.0067	0.0132	0.0147	0.0132	0.0342
	Testing	0.6153 ±	0.8670 ±	0.7188 ±	0.8287 ±	0.7416 ±	0.9238 ±	0.9140 ±	0.7262 ±	0.8801 ±	0.8537 ±
SP-TSK-RA1	ACC	0.0391	0.0263	0.0574	0.0195	0.0913	0.0110	0.0323	0.0403	0.0127	0.0551
	Training	0.7031	0.8561	0.7322	0.7993	0.7036	0.9184	0.9025	0.6976	0.8680	0.8811
	ACC	±0.1057	±0.0060	±0.0344	±0.0305	±0.0255	±0.0044	±0.0152	±0.0347	±0.0138	±0.0314
SP-TSK-RA2	Testing	0.5447	0.8542	0.6900	0.7836	0.6607	0.9160	0.8712	0.7152	0.8668	0.8244
	ACC	±0.0533	±0.0351	±0.0750	±0.0279	±0.0794	±0.0041	±0.0179	±0.0197	±0.0106	±0.0539
	Training	0.5086	0.8498	0.7031	0.7580	0.7248	0.9202	0.8482	0.6123	0.8534	0.8995
SP-TSK-RA2	ACC	±0.1084	±0.0073	±0.0170	±0.0151	±0.0322	±0.0037	±0.0174	±0.0041	±0.0101	±0.0227
	Testing	0.4513	0.8087	0.6550	0.7629	0.7096	0.9095	0.8290	0.6124	0.8373	0.7719
	ACC	±0.0614	±0.0273	±0.0534	±0.0268	±0.0964	±0.0065	±0.0273	±0.0165	±0.0181	±0.0567

5. Conclusions

In this paper, a novel self-paced Takagi-Sugeno-Kang (TSK) fuzzy classifier via the robust antecedents of all fuzzy rules called SP-TSK-RA is developed to achieve enhanced generalization capability and robustness on datasets containing ambiguous samples. Specifically, to achieve stable antecedents of the fuzzy rules, SP-TSK-RA learns the antecedent centers using ensemble clustering methods. To enhance generalization capability for handling datasets containing ambiguous samples, SP-TSK-RA designs a novel objective function for learning the consequents of fuzzy rules by adaptively identifying these ambiguous samples. Additionally, the optimal consequents of fuzzy rules can be learned using alternating optimization method and fewer major rules can also be obtained. Empirical outcomes on all utilized datasets demonstrate that SP-TSK-RA achieves enhanced and robust generalization capability and better interpretability (i.e., fewer fuzzy rules) compared to all adopted comparative methods.

The further work of SP-TSK-RA can be summarized as follows: (1) although SP-TSK-RA achieves enhanced generalization capability on most adopted datasets, developing a robust or enhanced fuzzy classifier by a deep or wide learning strategy for handling high-dimensional datasets is an interesting research direction. (2) SP-TSK-RA exhibits a high computational cost due to its antecedent construction employing a fuzzy ensemble clustering; hence, other optimization techniques (e.g., parallel computing) should be further developed to speed up the optimization process. (3) Additionally, although the used shortest-distance method is computationally efficient, developing new strategies to efficiently transform the continuous output of the TSK fuzzy classifier to discrete class labels is also an interesting topic.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Grok 4 in order to improve language. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Funding

This work was supported in part by the National Natural Science Foundation of China under Grant 62306126, in part by Jiangsu Province Youth Science and Technology Talent Support Project under Grant JSTJ2024283, in part by the Natural Science Foundation of Jiangsu Province under Grant BK20220621, in part by the Wuxi Science and Technology Development Fund Project under Grant K20231006.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ins.2026.123902>.

Data availability

- A. The data used in this research are available in a GitHub repository as described in the article.
- B. [datasets for SP-TSK-RA fuzzy classifier \(Reference data\)](#) (GitHub)

References

- [1] B. Qin, F.-L. Chung, S. Wang, Biologically plausible fuzzy-knowledge-out and its induced wide learning of interpretable TSK fuzzy classifiers, *IEEE Trans. Fuzzy Syst.* 28 (2020) 1276–1290.
- [2] Y. Qin, L. Wang, The optimization of TSK regression model based on error patch learning algorithm, *Int. J. Fuzzy Syst.* 27 (2025) 2258–2271.
- [3] T. Cover, P. Hart, Nearest neighbor pattern classification, *IEEE Trans. Inf. Theory* 13 (1967) 21–27.
- [4] Y. Zhang, H. Ishibuchi, S. Wang, Deep Takagi-Sugeno-Kang fuzzy classifier with shared linguistic fuzzy rules, *IEEE Trans. Fuzzy Syst.* 26 (2018) 1535–1549.
- [5] Y. Cui, D. Wu, J. Huang, Optimize TSK fuzzy systems for classification problems: minibatch gradient descent with uniform regularization and batch normalization, *IEEE Trans. Fuzzy Syst.* 28 (2020) 3065–3075.
- [6] J. Cao, T. Zhou, S. Lam, Y. Zhang, J. Zhang, X. Fan, D. Sun, J. Cai, A lightweight TSK fuzzy classifier with quantitative equivalent fuzzy rules via adaptive weighting, *IEEE Trans. Fuzzy Syst.* 33 (2025) 2805–2817.
- [7] E. Zhou, C.-M. Vong, Y. Nojima, S. Wang, A fully interpretable first-order TSK fuzzy system and its training with negative entropic and rule-stability-based regularization, *IEEE Trans. Fuzzy Syst.* 31 (2023) 2305–2319.

- [8] Q. Lou, W. Sun, W. Zhang, Z. Deng, K.S. Choi, S. Wang, Rules-based heterogeneous feature transfer learning using fuzzy inference, *IEEE Trans. Fuzzy Syst.* 32 (2024) 306–321.
- [9] A. Li, Z. Deng, W. Zhang, Z. Xiao, K.-S. Choi, Y. Liu, S. Hu, S. Wang, Multiview transfer representation learning with TSK fuzzy system for EEG epilepsy detection, *IEEE Trans. Fuzzy Syst.* 32 (2024) 38–52.
- [10] F. Guo, J. Liu, M. Li, T. Huang, Y. Zhang, D. Li, H. Zhou, A concise TSK fuzzy ensemble classifier integrating dropout and bagging for high-dimensional problems, *IEEE Trans. Fuzzy Syst.* 30 (2022) 3176–3190.
- [11] E. Zhou, C.M. Vong, Y. Nojima, S. Wang, Internally and generatively decorrelated ensemble of first-order Takagi–Sugeno–Kang fuzzy regressors with quintuply diversity guarantee, *IEEE Trans. Fuzzy Syst.* 32 (2024) 1288–1302.
- [12] Y. Li, F.L. Chung, Y. Nojima, S. Wang, Fully interpretable Gaussian centralized TSK fuzzy classifier from probabilistic perspective: concepts, output-stability-based learning, and ensemble, *IEEE Trans. Fuzzy Syst.* 33 (2025) 3967–3980.
- [13] E. Zhou, F.-L. Chung, S. Wang, Multiview fully interpretable TSK fuzzy classifier enhanced by multiview accompanying GMMs, *IEEE Trans. Fuzzy Syst.* 32 (2024) 6288–6302.
- [14] G. Suhan, C.M. Vong, P.K. Wong, S. Wang, Fast training of adversarial deep fuzzy classifier by downsizing fuzzy rules with gradient guided learning, *IEEE Trans. Fuzzy Syst.* 30 (2022) 1967–1980.
- [15] M. Su, R.R. Rhinehart, A generalized TSK model with a novel rule antecedent structure: structure identification and parameter estimation, *Comput. Chem. Eng.* 34 (2010) 1199–1219.
- [16] M. Sugeno, T. Yasukawa, A fuzzy-logic-based approach to qualitative modeling, *IEEE Trans. Fuzzy Syst.* 1 (1993) 7.
- [17] B. Qin, F.-L. Chung, S. Wang, KAT: a knowledge adversarial training method for zero-order Takagi–Sugeno–Kang fuzzy classifiers, *IEEE Trans. Cybern.* 52 (2022) 6857–6871.
- [18] B. Qin, F.-L. Chung, Y. Nojima, H. Ishibuchi, S. Wang, Fuzzy rule dropout with dynamic compensation for wide learning algorithm of TSK fuzzy classifier, *Appl. Soft Comput.* 127 (2022) 109410.
- [19] H. Ouifak, A. Idri, On the performance and interpretability of Mamdani and Takagi–Sugeno–Kang based neuro-fuzzy systems for medical diagnosis, *Sci. African* 20 (2023) e01610.
- [20] L. Pan, C. Gao, J. Zhou, G. Chen, X. Yue, Three-way decision-based Takagi–Sugeno–Kang fuzzy classifier for partially labeled data, *Appl. Soft Comput.* 164 (2024) 112010.
- [21] O. Cordon, F. Herrera, F. Hoffmann, L. Magdalena, Genetic fuzzy systems: evolutionary tuning and learning of fuzzy knowledge bases, World Scientific, 2001.
- [22] T. Takagi, M. Sugeno, Fuzzy identification of systems and its applications to modeling and control, *IEEE Trans. Syst. Man Cybern.* 15 (1985) 116–132.
- [23] M. Ma, J. Cai, Synchronization of Lagrangian networks with a directed graph via aperiodically intermittent pinning control, *Neurocomputing* 313 (2018) 175–183.
- [24] R. Xie, S. Wang, A wide interpretable Gaussian Takagi–Sugeno–Kang fuzzy classifier and its incremental learning, *Knowl.-Based Syst.* 241 (2022) 108203.
- [25] Y. Jiang, Z. Deng, K.-S. Choi, F.-L. Chung, S. Wang, A novel multi-task TSK fuzzy classifier and its enhanced version for labeling-risk-aware multi-task classification, *Inf. Sci.* 357 (2016) 39–60.
- [26] A.L.N. Fred, A.K. Jain, Combining multiple clusterings using evidence accumulation, *IEEE Trans. Pattern Anal. Mach. Intell.* 27 (2005) 835–850.
- [27] Z. Bian, J. Qu, J. Zhou, Z. Jiang, S. Wang, Weighted adaptively ensemble clustering method based on fuzzy Co-association matrix, *Inf. Fusion* 103 (2024) 102099.
- [28] P. Zhou, R. Li, Z. Ling, L. Du, X. Liu, Fair clustering ensemble with equal cluster capacity, *IEEE Trans. Pattern Anal. Mach. Intell.* 47 (2025) 1729–1746.
- [29] J.-S.-R. Jang, ANFIS: adaptive-network-based fuzzy inference system, *IEEE Trans. Syst. Man Cybern.* 23 (1993) 665–685.
- [30] M.P. Kumar, B. Packer, D. Koller, Self-paced learning for latent variable models, in: *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, Vancouver, BC, Canada, Dec. 2010, pp. 1189–1197.
- [31] Z. Bian, J. Zhang, F.-L. Chung, S. Wang, Residual sketch learning for a feature-importance-based and linguistically interpretable ensemble classifier, *IEEE Trans. Neural Netw. Learn. Syst.* 35 (2024) 10461–10474.
- [32] T. Zhou, H. Ishibuchi, S. Wang, Stacked-structure-based hierarchical Takagi–Sugeno–Kang fuzzy classification through feature augmentation, *IEEE Trans. Emerg. Top. Comput. Intell.* 1 (2017) 421–436.
- [33] H. Chen, J. Xie, X. Han, TSK-BN-2AWKD: a TSK fuzzy system optimization method for classification problems, *Int. J. Intell. Comput. Cybern.* 18 (2025) 397–417.
- [34] S.V. Aruna Kumar, E. Yaghoubi, H. Proença, A fuzzy consensus clustering algorithm for MRI brain tissue segmentation, *Appl. Sci.* 12 (2022) 7385.
- [35] L. Jiang, D. Meng, Q. Zhao, S. Shan, A.G. Hauptmann, Self-paced curriculum learning, in: *Proc. AAAI Conf. Artif. Intell.*, Austin, TX, USA, Jan. 2015, pp. 2694–2700.
- [36] Y. Tian, Y. Zhang, A comprehensive survey on regularization strategies in machine learning, *Inf. Fusion* 80 (2022) 146–166.
- [37] P. Zhou, L. Du, X. Liu, Z. Ling, X. Ji, X. Li, Y.-D. Shen, Partial clustering ensemble, *IEEE Trans. Knowl. Data Eng.* 36 (2024) 2096–2109.
- [38] M. Cococcioni, B. Lazzerini, F. Marcelloni, A TSK fuzzy model for combining outputs of multiple classifiers, in: *Proc. IEEE Int. Conf. Fuzzy Inf. Process. (NAFIPS)*, Banff, AB, Canada, Jun. 2004, pp. 871–876.
- [39] Z. Deng, K.S. Choi, F.L. Chung, S. Wang, Scalable TSK fuzzy modeling for very large datasets using minimal-enclosing-ball approximation, *IEEE Trans. Fuzzy Syst.* 19 (2011) 210–226.
- [40] X. Gu, F.-L. Chung, S. Wang, Bayesian Takagi–Sugeno–Kang fuzzy classifier, *IEEE Trans. Fuzzy Syst.* 25 (2017) 1655–1671.
- [41] A.J. Smola, B. Schölkopf, A tutorial on support vector regression, *Stat. Comput.* 14 (2004) 199–222.
- [42] K. Zhou, C. Fu, S. Yang, Fuzziness parameter selection in fuzzy c-means: the perspective of cluster validation, *Sci. China Inf. Sci.* 57 (2014) 1–8.
- [43] H. Wang, D. Xu, Parameter selection method for support vector regression based on adaptive fusion of the mixed Kernel function, *J. Control. Sci. Eng.* 2017 (2017) 3614790.
- [44] J. Demsar, Statistical comparisons of classifiers over multiple data sets, *J. Mach. Learn. Res.* 7 (2006) 1–30.
- [45] H.W. Kuhn, The Hungarian method for the assignment problem, *Naval Research Logistics Quarterly* 2 (1955) 83–97.



Jia Qu has received Ph.D. degree in control theory and control engineering from China University of Mining and Technology, Xuzhou, China and is currently an assistant professor of the School of Computer Science and Artificial Intelligence & Aliyun School of Big Data, Changzhou University, Changzhou, China. Her current research interests include machine learning, computational Bioinformatics, data fusion, and data mining.



Wulin Yang is currently working toward the master's degree with the School of Computer Science and Artificial Intelligence & Aliyun School of Big Data, Changzhou University, Changzhou, China. His research interests include machine-learning, ensemble clustering, and-fuzzy systems.



Zekang Bian has received Ph.D. degree in Jiangnan University and is currently an associate professor of the School of AI & Computer Science, Jiangnan University, Wuxi, China. And he is also served for PRC Ministry of Education Engineering Research Center of Intelligent Technology for Healthcare, Wuxi, Jiangsu 214122, China. His research interests include fuzzy artificial intelligence, pattern recognition, interpretable fuzzy systems, fuzzy/nonfuzzy clustering, ensemble clustering, ensemble learning, data fusion, and data mining.